



# Streaming platforms vs. theaters: Navigating the strategic choices of film distributors in channel selection<sup>☆</sup>

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## ABSTRACT

The increasing prevalence of streaming platforms is transforming the film distribution landscape, compelling distributors to reassess their channel strategies and consumers to modify their viewing habits. This transformation has garnered considerable attention in both the film industry and academia, especially concerning the determination of optimal release channels for films. Therefore, this study explores the strategic considerations of film distributors through the development of a game-theoretic model that integrates the effects of market expansion, channel differentiation, and diminishing consumer utility on decision-making processes. This paper examines the profitability and social welfare of the film distributor, streaming platform, and theater under different strategies. Through a comprehensive analysis, this study aims to reveal the critical factors that influence the profits and social welfare of all parties under different strategies. Our findings suggest that under specific conditions, prioritizing the theatrical release of films or releasing them exclusively on streaming platforms can lead to higher profitability for distributors. Additionally, we investigate the effects of delayed release times and channel differences on the equilibrium strategies of streaming platforms and theaters. Our analysis demonstrates that the hassle cost coefficient significantly influences the profitability of both streaming platforms and theaters across different strategies. Ultimately, we observe that prices and demand in both streaming and theatrical channels are affected by consumer utility loss due to delayed purchases. This study establishes a comprehensive theoretical framework to guide strategic decision-making for film distributors, streaming platforms, and theaters. Additionally, it offers valuable insights into the development of the film industry, serving as an essential reference for further research and strategic planning.

## 1. Introduction

The film industry holds significant standing as an integral component of national culture and social fabric, exerting a pivotal influence on the national economy. In 2023, the U.S. and Canada grossed about \$8.91 billion at the box office, well above the previous year's \$7.37 billion.<sup>1</sup> In China, in 2023, there were 54.915 billion RMB at the box office and 1.299 billion people watched films in theaters.<sup>2</sup> A stellar film has the potential to generate substantial revenues across the entire supply chain of the film industry, encompassing various stakeholders from production to distribution. As an exemplar of the film industry's prowess, Avatar, the most financially successful film of the 21st century, garnered over \$2.9 billion in global box office receipts.<sup>3</sup>

While distribution expenses constitute a smaller fraction of a film's overall investment compared to expenditures on actor compensation, production, and marketing, effective release channels are pivotal to a film's reputation and, ultimately, its financial performance. The distribution of a film is administered by a film distributor, serving as a crucial intermediary in the film industry's supply chain.

Film distributors, a crucial component in the dissemination of cinematic works, acquire rights from producers and determine the modalities for releasing films. Among the globally renowned distributors are Sony, Paramount, Universal, Disney and Warner Bros., who collectively maintain a significant market share in film distribution. The selection of release channels by film distributors is a matter of considerable significance for businesses and consumers. Presently, distributors typically

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<sup>1</sup> The information comes from <https://www.statista.com/statistics/187069/north-american-box-office-gross-revenue-since-1980/> (available at March 10, 2024).

<sup>2</sup> The information comes from <https://mp.weixin.qq.com/s/Xc31JXELavGngASTo4tDwg> (available at March 19, 2024).

<sup>3</sup> See <https://www.the-numbers.com/box-office-records/worldwide/all-movies/cumulative/all-time> (available at March 16, 2024).

have two primary options for releasing films: theaters or streaming platforms. The theatrical release of films is a time-honored practice, with the history of movie theaters dating back to the 19th century. Over time, cinemas have evolved beyond their initial function as mere venues for film screenings, becoming hubs for social and cultural interactions. Furthermore, many innovative technologies such as 3D theaters, IMAX big screen technology, VR technology, and Dolby atmos, which enhance the consumer experience, have been incorporated. Consequently, for a considerable duration, the theater channel has remained a preferred method for film distributors.

However, we can observe that: in 2020, the United States had 5798 movie theaters, a figure that is comparable to the 5773 theaters recorded a decade earlier.<sup>4</sup> Similar observations have been made in other countries and regions. Similar trends have been observed in other countries and regions, indicating that cinemas may face profitability challenges. This is primarily due to the increasing popularity of streaming services and rising movie ticket prices, leading many consumers to prefer watching films via streaming platforms. Streaming platforms now play a pivotal role in the film industry, both presently and prospectively. For instance, as of the fourth quarter of 2023, Netflix, a leading streaming service, reported approximately 260 million paying subscribers worldwide, marking an increase of over 13 million from the previous quarter.<sup>5</sup> This exemplifies how technological advancements and the increased accessibility of platforms have expedited the growth of streaming services. The digital distribution of films through streaming platforms is becoming increasingly prevalent, making them a viable option for film distributors.

Considering the coexistence of theater and streaming distribution and diverse audience preferences in the film industry, our study focuses on film distributors' strategic selection of new-film release channels. This choice, along with pricing strategies, raises several compelling research questions: (1) When a film distributor releases a new film to the market, what constitutes the optimal release channel strategy? (2) How do the length of the release delay and channel differences influence the interplay between different release channel strategies? (3) How do the profits of streaming platform and theater vary under diverse strategies? (4) What is the impact on pricing and demand dynamics when a new film is released under different strategies? Despite its significant practical implications and intricate nature, the current academic literature lacks a comprehensive study of such a particular problem. In this work, we leverage the Hotelling model to formulate a game-theoretic framework that delves into the strategic decisions regarding release channels adopted by film distributors.

In this study, we synthesize the multifaceted effects of market expansion, channel differences, and the declining consumer utility on the strategic decisions made by managers. Our findings have led us to the following key conclusions. First, we find that the channel strategy employed by film distributors is influenced by a plethora of factors. However, in the majority of cases, our findings indicate that the theater-first strategy or the streaming platform-only channel strategy performs better. Furthermore, the delayed release times of films often result in adverse effects on the later-releasing channel, thereby benefiting the initial-releasing channel financially. Additionally, the channel differences between different channels tend to favor the streaming platform channel, ultimately diminishing the profitability of the theater channel. Subsequently, variations in the profitability of the streaming platform and the theater are influenced by the coefficient of hassle cost. This factor results in differing rates of profit change and exhibits varying relationships within distinct thresholds. Finally, the pricing and demand

dynamics for newly released films, which are launched under varying strategies, are significantly influenced by consumers' perceived utility loss stemming from delayed purchases.

The remainder of this study is organized as follows. In Section 2, a categorical review of the relevant literature is presented. Section 3 introduces the model framework, delineates key parameters, and conducts an analysis of pricing decisions under four scenarios. Section 4 undertakes a comparative analysis of the strategies. Section 5 conducts a welfare analysis. Moreover, in Section 6, we present extended research in three aspects: consumer heterogeneity, Nash bargaining, and the dynamic influence of electronic word-of-mouth. Section 7 summarizes the research conclusions and managerial implications. Detailed proofs of the propositions and corollaries are provided in the Appendix for reference.

## 2. Literature review

Our study intersects with three prevalent strands of research: content products such as films, release channels for films, and streaming platforms. Next, we will delve into detailed discussions of these areas.

### 2.1. Content products

Over recent decades, the importance of content products has risen to prominence in the academic field (Yang et al., 2021; Zhou et al., 2025a). Utilizing daily ticket prices and attendance data for theaters, Ho et al. (2018) empirically investigated the potential benefits of differentiated movie-specific pricing in theaters and compared it with the increasingly prevalent two-tiered (2D/3D) uniform pricing strategy, relative to a fully uniform pricing approach where all movies are charged at the same rate. Based on Rogers' Diffusion of Innovations theory, Baek et al. (2017) examined the impact of electronic word-of-mouth (eWOM) across various types of social media on box office revenues at different stages of a film's screening cycle. Ullah et al. (2025) empirically examined the effects of positive and negative content on movie sales. They found that negative emotional content had a limited impact on sales, beyond which it actually boosted sales. The study further revealed that when the volume of electronic word-of-mouth was high, the impact of positive content gradually strengthened, whereas that of negative content exhibited diminishing returns. Suckfüll and Moellering (2015) analyzed the relationship between film trailers and receptive patterns using an empirical research approach through an online survey with 554 participants. The study found that using receptive mode as a consumer characteristic and intentionally distributing produced film trailers to target groups can increase the efficiency of film marketing campaigns. Taking film trailer in New Zealand as the subject of their study, Finsterwalder et al. (2012) examined the impact of consumer expectations on film content and quality through a qualitative analytical approach to research. Zhang et al. (2022) employed a dynamic analytical framework to investigate consumers' time-inconsistent preferences in digital content consumption and their strategic self-control behaviors. The study also evaluated the effects of pay-per-chapter pricing and monthly subscription models on both platform profitability and consumer welfare. In addition, Yang et al. (2023) examined collaboration models between digital content platforms and IP holders and found that platforms' selection of a collaboration model is influenced by factors such as peak demand, cost efficiency, and disparities in the capability to generate content-derived revenue between platforms and intellectual property (IP) holders.

### 2.2. Release channels for films

With the emergence of different release channels, numerous studies have focused on the competitive dynamics between release channels (Zhou et al., 2024, 2025b). Ahmed and Sinha (2016) examined the optimization of film release timing decisions using box office and DVD

<sup>4</sup> The information comes from <https://www.statista.com/statistics/188643/number-of-us-cinema-sites-since-1995/> (available at March 19, 2024).

<sup>5</sup> The information comes from <https://www.statista.com/statistics/250934/quarterly-number-of-netflix-streaming-subscribers-worldwide/> (available at March 19, 2024).

sales in the United States. The revenues of these two channels are jointly modeled through copula modeling approach to improve management decisions. It is found that delayed DVD release is still the best option for maximizing revenue in the DVD channel. [Duarte de Souza et al. \(2025\)](#) found that the introduction of Netflix streaming services is associated with a 14%–17% reduction in box office revenues, providing empirical evidence of cannibalization between streaming platform and theatrical channels. [Lehmann and Weinberg \(2000\)](#) estimated exponential sales curves for theater attendance and video rentals using data from 35 films, demonstrating how the sales parameters of the first channel (theaters) can be utilized to predict sales in the second channel (video rentals). From the perspective of a film distributor, the optimal release timing was determined based on the model and its estimated parameters. The study's findings indicate that releasing films earlier onto streaming platforms could lead to increased profits. [Brennan et al. \(2025\)](#) using a two-stage difference-in-difference analysis, the study found that when a film is made available on a secondary channel such as a streaming platform, both the number of theaters showing the film and its box office gross significantly decrease, and this negative impact continued to intensify as the film remains available on the secondary channel for a longer period. Furthermore, [Sharma et al. \(2023\)](#) analyzed the movie distributor's decision-making problem, and elaborated on how the optimal release time fluctuates based on platform attributes, film attributes, and market dynamics, offering essential guidelines for professionals involved in movie distribution on OTT platforms. [Wan et al. \(2020\)](#) examined the impact of O2O platforms on the theater industry by modeling general duopoly competition. They found that the degree of differentiation of competing theaters decreases with the addition of O2O platforms as either of the two factors, total market size or the proportion of audience preferring ticket services through O2O platforms, increases.

To further elucidate the practical implications of the theoretical analysis, the researchers conducted an empirical study, yielding a series of significant and insightful conclusions. [Díaz et al. \(2017\)](#) evaluated the model using partial least squares (PLS) in order to analyze the differences between online and offline consumer behavior. A multi-group analysis compared consumers who purchased tickets online with those who purchased tickets at the box office. Relevant theoretical and empirical perspectives are also considered, providing important recommendations for shopping center managers, movie theaters, and intermediaries in the industry. [Mukherjee and Kadiyali \(2018\)](#) constructed a structural model of dynamic competition wherein studios must balance the trade-offs between waiting for high-demand periods, the diminishing freshness of content, and intensified competitive crowding. The findings of the study align with conservative management strategies: studio executives opt to forego DVD revenues from smaller films in order to prioritize and ensure the DVD success of larger, more significant releases. Furthermore, [Somlo et al. \(2011\)](#) initially developed a method to predict theater-level box office receipts for a given film based on its attributes and theater characteristics. Film distributors subsequently utilize these predictions to determine optimal screening locations. The distributor's location selection problem is formulated as an integer programming optimization model aimed at maximizing profits by selecting the most suitable theaters. This approach is validated using real-world box office data, demonstrating its potential to significantly enhance distributor profits. To empirically analyze the interaction between purchase and rental markets, [Rao \(2015\)](#) developed a preference-measurement instrument that infers consumers' intertemporal preferences solely from their current choices. These inferred preferences were subsequently utilized to derive a dynamic equilibrium between consumers and firms. Ultimately, a number of interesting conclusions are obtained.

### 2.3. Streaming platforms

There is a large body of literature on streaming platforms in the context of rapid advances in information technology, aiming to emphasize the key role that these platforms play in the supply chain of content products. [Bender et al. \(2021\)](#) investigated the conditions under which artists can monetize their music releases through streaming platforms by building a game-theoretic model. Taking into account the heterogeneity of the artist population, the value users place on the diversity offered by the platform and the revenue sharing rules, a number of interesting columns are obtained. [Bergantiños and Moreno-Ternero \(2025\)](#) analyzed the allocation of subscription revenues among content creators on music streaming platforms from an axiomatic and game-theoretic perspective. It found that two widely used methods each had their theoretical foundations. These methods were distribution based on total streams and distribution based on each user's individual listening behavior. On this basis, the study further proposed two new allocation schemes to better align the interests of artists, users, and platforms. [Xing et al. \(2023\)](#) focused on a supply chain comprising a video service platform and a video service provider (VSSC). Based on data-driven marketing (DDM), the study found that the optimal strategies and profits of VSSC members were jointly influenced by the basic market size, fan preferences, and DDM efficiency. [Fang et al. \(2023\)](#) developed a hotelling model to study the optimal content differentiation policies of streaming platforms. Their model, which is based on maximizing customer engagement and understanding customer tastes while considering consumer utility and production costs, produced insightful results on media companies' content development policies. [Meyn et al. \(2023\)](#) investigated the side effects, both supply and demand, of changing the currently used pro-rata pay policy to a user-centered one, as well as the resulting change in profits. Using personal data from 3326 participants and data from the Spotify API, they empirically quantified the monetary impact of this change on the music industry and examined the impact of business practices on users, platforms and artists in the music streaming industry. [Guo et al. \(2024\)](#) examined an interesting emerging service strategy in which content platforms that are competitors to each other jointly offer membership bundles that allow consumers to access their platforms at a lower price. They argued that this approach transforms platform relationships from competitive to cooperative dynamics. By employing a game-theoretic model, they illustrated how membership bundling influences content distribution and access outcomes, along with the associated benefits. Additionally, they conduct several extensions to validate the robustness of their findings. [Yu et al. \(2025\)](#) investigated the factors that shape manufacturers' decisions to adopt live-streaming channels and how competitive dynamics influence those choices. [Zhang and Hua \(2025\)](#) investigated the strategic decision of online video platforms regarding original content investment and business model selection. It found that the production cost of original content is a key determinant. Furthermore, allowing free users limited access to original content can simultaneously expand the engagement of both subscriber and advertiser, thereby enhancing the streaming platform's dual revenue streams. In addition, [Bender et al. \(2022\)](#) investigated the strategies of established content providers selling digital downloads in response to emerging streaming technologies. The study revealed that streaming can function both as a substitute and a complement to digital content sales, with the degree of this relationship contingent upon the intensity of industry competition and the demographic composition of the consumer base.

### 2.4. Summary of differences from the previous literature

In summary, this paper differs from existing studies in the following three aspects. First, in the area of content products, only a limited number of studies have investigated the release of films as content products. To fill this research gap, this paper explores the strategic decisions

of film distributors by taking into account the distinct characteristics of various distribution channels and the inherent attributes of content products. Second, in the area of film release channels, in contrast to the aforementioned studies, our research conducts an in-depth analysis of potential film release channels, investigates the channel selection issues faced by film distributors, and presents valuable insights into film pricing strategies by incorporating user and platform characteristics. Third, in the area of streaming platforms, although some studies have examined the pricing strategies of streaming platforms, few have analyzed their pricing strategies of streaming platforms in the presence of competition from other types of channels. This study argues that streaming platforms attract consumers through content quality and additional services, considers the market expansion effect of streaming platforms, and analyzes the profits of streaming platforms, theaters, and film distributors. Together, these efforts contribute new theoretical insights to the literature on platform strategy and digital markets.

### 3. The model

In the context of film distribution, when a distributor aims to release a new film, the distributor is faced with a pivotal decision: whether to release the film through a streaming platform or a theater. This section delves into the competitive landscape between these two release channels and examines the strategic choice of film distribution by developing a game-theoretic model. For convenience, in the rest of the paper,  $d$ ,  $s$ , and  $t$  are used to denote the film distributor, the streaming platform, and the theater, respectively.

The traditional distribution of new films via the theater channel remains a prevalent practice. When a film distributor opts for a theater channel, consumers are expected to pay a ticket price of  $p_t$  to the theater in order to obtain access to the film and the theater is required to pay the film distributor based on the royalty rate  $\gamma$ . In the era of rapid advancements in information technology, the dissemination of new films via streaming platforms has emerged as a ubiquitous phenomenon in recent years. When a film distributor elects to release a new film via a streaming platform, consumers gain access to the film for a designated duration by subscribing to the platform, with a subscription price set at  $p_s$ . Upon subscribing to the streaming platform, consumers not only gain access to the new film but also enjoy complimentary viewing of other content and value-added services. Drawing from prior research (Yang et al., 2021, 2023), this paper employs the parameter  $\lambda$  to quantify the positive utility that the streaming platform's additional resources provide to its subscribers. Similarly, the streaming platform needs to remunerate the film distributor based on the royalty rate  $\beta$ . In our model, the parameters  $\gamma$  and  $\beta$  are considered exogenous, as the royalty rates for the two channels are based on industry consensus, and individual contract negotiations do not induce alterations in these rates.

Upon the release of a new film, whether via a streaming platform or theater, consumers formulate a perception of the film's quality based on its publicity, deriving a perceived utility  $v > 0$ . This perceived utility,  $v$ , serves as a composite metric encompassing various quality aspects of the film, including its theme, cast, directors, and celebrities. Notably, variations in consumers' perceived utility arise when film distributors opt for distinct release channels for their new films. In this study, we assume that consumers perceive utility as  $v$  when the film distributor chooses to release the new film through the theater, and as  $\theta v$  when it goes through the streaming platform.  $\theta$  represents a discount factor that reflects the disparity between the two channels. This assumption is made because we postulate that the viewing experience in theaters is more specialized. This specialization stems not only from differences in screen size and audio quality but also from the non-utilitarian advantages associated with theatrical viewing. These advantages include the social atmosphere, the immersive experience offered by technologies such as IMAX and Dolby Atmos, and potential physical souvenirs. As a result, consumers who opt to watch a film in a theater generally

enjoy a better experience than those who watch the same film through streaming platforms.

Similar to Jung et al. (2019) and Bakos and Halaburda (2020), we use the Hotelling model to capture the differences and competition between the two channels. We assume that consumers are uniformly distributed in Hotelling linear city to reflect the differences in their preferences. Therefore, the location of the consumer, denoted by  $x$ , obeys a uniform distribution on  $[0, 1]$ . In a linear market, the streaming platform and the theater are located at the two ends of the Hotelling line. Specifically, the streaming platform and the theater are located at 0 and 1. Consumers incur different hassle costs when spending in any channel, reflecting their overall attitudes towards non-price factors (convenience, user service, level of technology). For example, in the theater, consumers may need to spend time traveling to the theater, standing in line to purchase tickets, etc.; on the streaming platform, consumers may need to spend time searching, evaluating, and paying online. In our study, we use the parameter  $k$  to denote the user's hassle cost. Thus, a consumer located at  $x$  on the Hotelling line incurs a hassle cost  $kx$  when choosing the streaming platform and a hassle cost  $k(1-x)$  when choosing the theater.

Within the context of film distribution, a distributor possesses various strategic options for releasing their film. In this paper, we delve into four distinct release strategies: simultaneous release, theater first, release only through theater, and release only through streaming platform. Each of these strategies is characterized by a unique sequence of decisions. Fig. 1 depicts the decision sequence under each strategy. It elucidates the order in which the streaming platform and the theater determine their prices, along with the information accessible at each decision-making point. Subsequently, we formulate and solve analytical models pertaining to each of these four strategies.

#### 3.1. Strategy ST

First, we consider the situation where the streaming platform and the theater release the new film at the same time (referred to as strategy ST). When a film distributor chooses to release a new film through the ST strategy, the streaming platform and the theater simultaneously determine their own price  $p_s^{ST}$  and  $p_t^{ST}$ . Consumers can then choose to consume from either the streaming platform or the theater. The utility functions obtained are  $U_s^{ST} = \theta(v + \lambda) - p_s^{ST} - kx$  and  $U_t^{ST} = v - p_t^{ST} - k(1-x)$ , respectively.

Each consumer purchases a unit of the available product that gives him the maximum positive utility, and by comparing the utility of the two channels, the consumer ultimately chooses the channel that has positive and greater utility. By comparing the utility functions  $U_s^{ST}$  and  $U_t^{ST}$ , we get that the consumer located at  $\bar{x}$  obtain the same utility from both channels. This consumer is  $\bar{x}$  away from the streaming platform and  $1 - \bar{x}$  away from the theater, with representing the no-difference point.

Therefore, the demand functions for the streaming platform and the theater under strategy ST are:  $D_s^{ST} = \frac{(1+\alpha)(\theta(v+\lambda)-p_s^{ST}-v+p_t^{ST}+k)}{2k}$  and  $D_t^{ST} = 1 - \frac{\theta(v+\lambda)-p_s^{ST}-v+p_t^{ST}+k}{2k}$ , where parameter  $\alpha$  represents the strength of the market expansion effect of the streaming platform. This is because the streaming platform is more accessible to reach out to consumers than the theater.

Thus, under strategy ST, profits for the streaming platform, theater, and film distributor are:

$$\pi_s^{ST} = (1-\beta) \frac{(1+\alpha)(\theta(v+\lambda)-p_s^{ST}-v+p_t^{ST}+k)}{2k} p_s^{ST}, \quad (1)$$

$$\pi_t^{ST} = (1-\gamma) \left(1 - \frac{\theta(v+\lambda)-p_s^{ST}-v+p_t^{ST}+k}{2k}\right) p_t^{ST}, \quad (2)$$

and

$$\begin{aligned} \pi_d^{ST} = & \beta \frac{(1+\alpha)(\theta(v+\lambda)-p_s^{ST}-v+p_t^{ST}+k)}{2k} p_s^{ST} \\ & + \gamma \left(1 - \frac{\theta(v+\lambda)-p_s^{ST}-v+p_t^{ST}+k}{2k}\right) p_t^{ST}. \end{aligned} \quad (3)$$

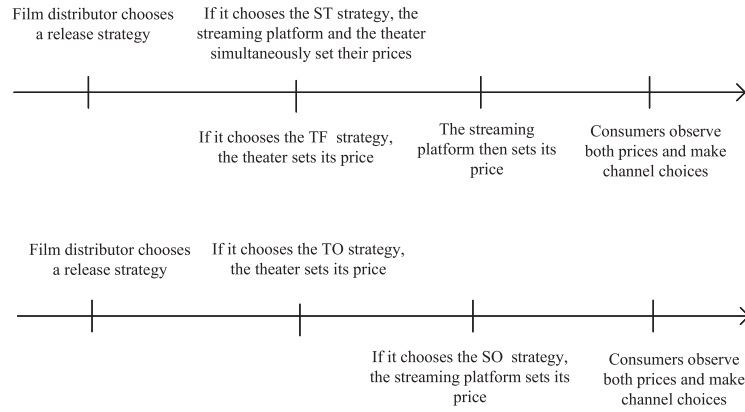


Fig. 1. Timing of events.

### 3.2. Strategy TF

We next analyze the situation where there is a time lag between the releases of the two channels. First, we consider the strategy in which the film distributor prioritizes theatrical release followed by streaming platform release (referred to as strategy TF). Under strategy TF, the theater first determines its price  $p_t^{TF}$ , and subsequently, the streaming platform decides its price  $p_s^{TF}$ . That is to say, only  $p_t^{TF}$  is announced at theatrical release, while  $p_s^{TF}$  is announced post-theatrical release, after the streaming platform observes the realized demand. Since films are similar to fashion items, digital products and seasonal products, the assumption that their valuation declines over time during sales seems common. Therefore, the novelty of the film decreases when the streaming platform releases the film later, and the perceived utility of consumers in the streaming channel becomes  $\theta e^{-b\tau}v$ . Here,  $v e^{-b\tau}$  represents consumer evaluations of new film over time  $\tau$ , and the parameter  $b$  reflects the extent to which the quality of the product has survived the test of time (in the analysis in Section 4, we adopt the notation  $\phi$  to represent  $e^{-b\tau}$  in order to maintain consistency and enhance clarity within the discourse). We acknowledge that institutional constraints in the real world, such as government-mandated theatrical windowing periods and content review procedures, exert a significant influence on distributors' channel strategies. In certain markets (e.g., China), these factors may even be decisive. To handle these influences in a tractable and generalizable way, our model captures their ultimate market effects via two key parameters:  $\phi$  and  $\theta$ . Specifically,  $\phi$  captures the overall utility loss resulting from delayed releases, effectively aggregating the impacts of various time-based constraints. Similarly,  $\theta$  reflects not only the inherent experiential disparities between channels but also the influence of regulatory intensity and content standards on the final product accessible on each platform. Hence, utility functions under strategy TF are  $U_s^{TF} = \theta(e^{-b\tau}v + \lambda) - p_s^{TF} - kx$  and  $U_t^{TF} = v - p_t^{TF} - k(1-x)$ .

Similarly, we obtain the demand functions under strategy TF as  $D_s^{TF} = \frac{(1+\alpha)(\theta(e^{-b\tau}v + \lambda) - p_s^{TF} - v + p_t^{TF} + k)}{2k}$  and  $D_t^{TF} = 1 - \frac{\theta(e^{-b\tau}v + \lambda) - p_s^{TF} - v + p_t^{TF} + k}{2k}$ . Thus, under strategy TF, profits for the streaming platform, theater, and film distributor are

$$\pi_s^{TF} = (1 - \beta) \frac{(1 + \alpha)(\theta(e^{-b\tau}v + \lambda) - p_s^{TF} - v + p_t^{TF} + k)}{2k} p_s^{TF}, \quad (4)$$

$$\pi_t^{TF} = (1 - \gamma) \left( 1 - \frac{\theta(e^{-b\tau}v + \lambda) - p_s^{TF} - v + p_t^{TF} + k}{2k} \right) p_t^{TF}, \quad (5)$$

and

$$\pi_d^{TF} = \beta \frac{(1 + \alpha)(\theta(e^{-b\tau}v + \lambda) - p_s^{TF} - v + p_t^{TF} + k)}{2k} p_s^{TF} + \gamma \left( 1 - \frac{\theta(e^{-b\tau}v + \lambda) - p_s^{TF} - v + p_t^{TF} + k}{2k} \right) p_t^{TF}. \quad (6)$$

### 3.3. Strategy TO

Next, we consider the case where the film distributor chooses to release the new film through a single channel. Under strategy TO, the new film is released only through the theater. At this point only the theater is deciding the price  $p_t^{TO}$ . In the market, consumers can only access the new film through the theater, where the consumer's utility function is  $U_t^{TO} = v - p_t^{TO} - k(1-x)$ . Letting  $U_t^{TO} = 0$ , we determine the location of the consumer who is indifferent between making a purchase and abstaining from purchasing. Therefore, the demand function is  $D_t^{TO} = \frac{v - p_t^{TO}}{k}$ .

Under strategy TO, selling the new film to consumers is the only source of profit for the theater and film distributor. Then their profits are

$$\pi_t^{TO} = \frac{(1 - \gamma)(v - p_t^{TO})}{k} p_t^{TO}, \quad (7)$$

and

$$\pi_d^{TO} = \frac{\gamma(v - p_t^{TO})}{k} p_t^{TO}. \quad (8)$$

### 3.4. Strategy SO

Under strategy SO, we consider the case where the new film is released only through the streaming platform. Similar to strategy TO, only the streaming platform decides the price  $p_s^{SO}$ , the consumer's utility function is  $U_s^{SO} = \theta(v + \lambda) - p_s^{SO} - kx$ .

Similarly, we can calculate the demand function at this point as  $D_s^{SO} = \frac{\theta(v + \lambda) - p_s^{SO}}{k}$ . The profits for the streaming platform and distributor are

$$\pi_s^{SO} = \frac{(1 - \beta)(\theta(v + \lambda) - p_s^{SO})}{k} p_s^{SO}, \quad (9)$$

and

$$\pi_d^{SO} = \frac{\beta(\theta(v + \lambda) - p_s^{SO})}{k} p_s^{SO}. \quad (10)$$

In order to find the optimal solutions for these four strategies, we use backward induction to calculate them. The equilibrium prices and profits are presented in Table 1, while the equilibrium demands are detailed in Table 2.

## 4. Analysis and discussion

In this section, we systematically analyze and discuss the equilibrium outcomes of these four scenarios. First, we compare the prices, demands, and profits under different release strategies. Subsequently, we conduct a numerical analysis to examine the effects of key parameters such as consumers' valuation and channel differences on the equilibrium decisions and illustrate the optimal strategy.

**Table 1**  
Equilibrium prices and profits under different strategies.

| Strategy | Participant        | Price                                                    | Profit                                                                                                                                               |
|----------|--------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| ST       | streaming platform | $\frac{3k - v + \theta v + \theta \lambda}{3}$           | $\frac{(1 + \alpha)(1 - \beta)(3k - v + \theta v + \theta \lambda)^2}{18k}$                                                                          |
|          | theater            | $\frac{3k + v - \theta v - \theta \lambda}{3}$           | $\frac{(1 - \gamma)(3k + v - \theta v - \theta \lambda)^2}{18k}$                                                                                     |
|          | distributor        | –                                                        | $\frac{\beta(1 + \alpha)(3k - v + \theta v + \theta \lambda)^2}{18k} + \frac{\gamma(3k + v - \theta v - \theta \lambda)^2}{18k}$                     |
| TF       | streaming platform | $\frac{5k - v + e^{-b\tau}\theta v + \theta \lambda}{4}$ | $\frac{(1 + \alpha)(1 - \beta)(5k - v + e^{-b\tau}\theta v + \theta \lambda)^2}{32k}$                                                                |
|          | theater            | $\frac{3k + v - e^{-b\tau}\theta v - \theta \lambda}{2}$ | $\frac{(1 - \gamma)(3k + v - e^{-b\tau}\theta v - \theta \lambda)^2}{16k}$                                                                           |
|          | distributor        | –                                                        | $\frac{\beta(1 + \alpha)(5k - v + e^{-b\tau}\theta v + \theta \lambda)^2}{32k} + \frac{\gamma(3k + v - e^{-b\tau}\theta v - \theta \lambda)^2}{16k}$ |
| TO       | streaming platform | –                                                        | –                                                                                                                                                    |
|          | theater            | $\frac{v}{2}$                                            | $\frac{(1 - \gamma)v^2}{4k}$                                                                                                                         |
|          | distributor        | –                                                        | $\frac{\gamma v^2}{4k}$                                                                                                                              |
| SO       | streaming platform | $\frac{\theta v + \theta \lambda}{2}$                    | $\frac{(1 + \alpha)(1 - \beta)\theta^2(v + \lambda)^2}{4k}$                                                                                          |
|          | theater            | –                                                        | –                                                                                                                                                    |
|          | distributor        | –                                                        | $\frac{\beta(1 + \alpha)\theta^2(v + \lambda)^2}{4k}$                                                                                                |

**Table 2**  
Equilibrium demands under different strategies.

| Strategy | Streaming platform                                                      | Theater                                                   |
|----------|-------------------------------------------------------------------------|-----------------------------------------------------------|
| ST       | $\frac{(1 + \alpha)(3k - v + \theta v + \theta \lambda)}{6k}$           | $\frac{3k + v - \theta v - \theta \lambda}{6k}$           |
| TF       | $\frac{(1 + \alpha)(5k - v + e^{-b\tau}\theta v + \theta \lambda)}{8k}$ | $\frac{3k + v - e^{-b\tau}\theta v - \theta \lambda}{8k}$ |
| TO       | –                                                                       | $\frac{v}{2k}$                                            |
| SO       | $\frac{(1 + \alpha)\theta(v + \lambda)}{2k}$                            | –                                                         |

#### 4.1. Analytical comparisons

Drawing upon the outcomes of these four scenarios, we present several salient findings in this subsection to offer insights into how the strategic decisions of film distributors influence optimal pricing and profitability in these contexts. First, by comparing the profits when releasing through dual channels, we derive the optimal strategy for the streaming platform and theater, as shown in [Proposition 1](#).

**Proposition 1.** *When the content distributor decides to release the film through dual channels, by analyzing the profits of the platform and the theater, we get the following results:*

(a) If  $0 < k < k_1$ , the streaming platform under the ST strategy gets higher profit; if  $k_1 < k < 1$ , the profit of streaming platform under the TF strategy is higher;

(b) If  $0 < k < k_2$ , the theater under the ST strategy gets higher profit; if  $k_2 < k < 1$ , the theater under the TF strategy gets higher profit.

[Proposition 1](#) demonstrates the relationship between equilibrium profits of streaming platform and theater under different strategies. We have the following findings:

First, when the hassle cost is small, the optimal strategy for the streaming platform is the ST strategy. This is because when consumer hassle costs are small, the streaming platform and theater become more substitutable for consumers. As a result, more consumers choose to compare the utility of the two before making a purchase. Therefore, both the platform and the theater have an incentive to compete with price cuts. This makes them less profitable. But profits declined at the slowest rate for streaming platform and theater under the ST strategy. Thus, when  $0 < k < k_1$  ( $0 < k < k_2$ ), the streaming platform (theater) is the most profitable under the ST strategy.

Contrarily, when  $k_1 < k < 1$ , the streaming platform under the TF strategy is more profitable. This is because when consumers' hassle

costs are high, there is less substitutability between the platform and theater for consumers, and more consumers become loyal customers of the streaming platform or theater. Therefore, the platform and the theater are incentivized to raise prices for higher returns. At this point, profits grow faster through a time-differentiated distribution strategy than the ST strategy. Thus, when factor  $k$  is larger, the ST strategy is no longer more profitable for streaming platforms and theaters. The analysis for theater is similar to that for the streaming platform, and the results are shown in [Proposition 1\(b\)](#).

Next, our analysis focuses on exploring the implications for film pricing and consumer demand when the film distributor opts for dual-channel distribution strategies.

**Proposition 2.** *By analyzing the prices and profits of dual-channel distribution, we have the following findings:*

(a) If  $0 < \phi < \phi_1$ , the demand of streaming platform under the ST strategy is higher; otherwise, it is higher under the TF strategy. If  $\phi_2 < \phi < 1$ , the demand of theater under the ST strategy is higher; otherwise, it is higher under the TF strategy. And,  $\phi_1 - \phi_2 = \frac{1}{3\theta} > 0$ ;

(b)  $0 < \phi < \phi_1$ , the price of streaming platform under the ST strategy is higher, and if  $\phi_3 < \phi < 1$  the price of theater under the ST strategy is higher.

When the difference in release time has a high impact on consumers, the streaming platform can get higher demand through the ST strategy. This is because, when the impact of time differences is large, the utility of the streaming platform that is released later through the TF strategy is significantly discounted, which makes it difficult to attract consumers. On the contrary, if the impact is small, then more consumers are attracted by the low price of the streaming platform, which leads to fewer consumers in theater. Therefore, the theater can obtain higher demand through the ST strategy when the release time difference has a small impact on consumers. In addition,  $\phi_1 > \phi_2$  suggests that under certain conditions, the streaming platform and the theater can obtain higher demand through the ST strategy.

Moreover, as illustrated in [Proposition 2\(b\)](#), when demand increases, both the streaming platform and the theater have an incentive to raise prices for higher profits.

Then, we study the response of the streaming platform and theater when the film distributor opts for a single-channel release and express this as [Proposition 3](#).

**Proposition 3.** *When the content distributor chooses to release through a single channel,*

(a) If  $0 < \theta < \frac{v}{v+\lambda}$ , the ticket price under the TO strategy is higher than the subscription price under the SO strategy, indicating,  $p_t^{TO*} > p_s^{SO*}$ ; otherwise,  $p_t^{TO*} < p_s^{SO*}$ ;

(b) If  $0 < \theta < \frac{v}{(1+\alpha)(v+\lambda)}$ , the demand for the TO strategy is higher than that for the SO strategy, as denoted by,  $D^{TO*} > D^{SO*}$ ; otherwise,  $D^{TO*} < D^{SO*}$ ;

(c) If  $\alpha < \frac{(1-\gamma)v^2}{(1-\beta)\theta^2(v+\lambda)^2} - 1$ , the profit of theater under the TO strategy are higher than that of streaming platform under the SO strategy, or else the profit of streaming platform under the SO strategy are higher.

**Proposition 3** indicates that the SO strategy has a low subscription price when there is a large difference between the two channels. This is because when consumers gain a large loss of utility by consuming through the streaming platform, a portion of users with high hassle costs choose not to make purchases, which leads to a loss of profits for the platform. In order to avoid this loss, the platforms have a motivation to offer a lower price to the consumer and thus get to recoup some of their losses. Thus, when  $0 < \theta < \frac{v}{v+\lambda}$ ,  $p_t^{TO*} > p_s^{SO*}$ . When the loss of utility gained by the consumer through the streaming platform is small ( $\frac{v}{v+\lambda} < \theta < 1$ ), the utility gained by the consumer from consuming on the streaming platform increases at this point, which causes more and more consumers to subscribe to the platform. Thus the platform manager has an incentive to increase the price to gain more profit. At this time,  $p_t^{TO*} < p_s^{SO*}$ . Similarly, we find that  $D^{TO*} > D^{SO*}$  when  $0 < \theta < \frac{v}{(1+\alpha)(v+\lambda)}$ .  $D^{TO*} < D^{SO*}$  when  $\frac{v}{(1+\alpha)(v+\lambda)} < \theta < 1$ . Therefore, for content distributors, when choosing a single-channel distribution strategy, care should be taken to identify the differences in the consumer experience that film productions bring to different channels. Higher prices, demands, and profits can be realized through rational strategy choices.

Besides, the size of the profit of the theater under the TO strategy and that of the streaming platform under the SO strategy are affected by the market expansion effect. When the expansion effect is small, the theater benefits from the more specialized equipment that gets more consumer preference and more profit under the TO strategy; when the expansion effect is large, the streaming platform benefits from a larger number of consumers and makes more profit under the SO strategy.

We next analyze the effect of delayed release times and inter-channel differences on equilibrium solutions, and the results are summarized in **Proposition 4**.

**Proposition 4.** Under different strategies, the effects of delayed release time and channel difference on the equilibrium decision are as follows:

(a) When the delayed release time becomes longer, the price, demand, and profit of the channel that releases first always increases; and there is a consequent reduction the channel that releases later;

(b) As the difference between the streaming platform channel and the theater channel become smaller, there is a corresponding increase in the streaming platform's price, demand, and profit; and that of the theater channel decreases.

**Proposition 4(a)** analyzes the reaction of the two channels when the delayed release time becomes longer. This result is intuitive. This is because when the delayed release time becomes longer, the film becomes less useful as a fashionable product that consumers can perceive. So there are fewer consumers choosing the delayed release channel and more choosing the early release channel. It also incentivizes streaming platform and theater to make strategic price adjustments. As a result, the late-release channel suffers a large loss of profit.

**Proposition 4(b)** examines the impact on both channels when the difference between the streaming platform channel and the theater channel changes. As the differences diminish, the streaming platforms are able to charge higher prices and make higher profits, and the theater channel is negatively impacted. Therefore, one of the conditions for streaming platforms and theaters to be highly profitable is to improve the customer's consumption experience, thus reducing or increasing channel variance.

## 4.2. Numerical analysis

Due to the complexity of the expressions, we perform a numerical analysis in this subsection to derive the characteristics of the optimal strategy features.

To ensure that the management insights obtained from the numerical analysis have a realistic foundation, this study empirically calibrated the key parameters using authoritative data. The parameter  $\theta = 0.6$  is calibrated by integrating the consumer willingness-to-pay survey data from the "China Film Market Report" and referring to the international pricing cases disclosed in Netflix's financial reports and the MPAA report.<sup>6</sup> The parameter  $\lambda = 0.7$  is determined based on the tiered pricing data of platforms such as IQIYI and Tencent Video, which quantifies the marginal utility of value-added services like 4 K quality and Dolby sound. The parameter  $k = 0.2$  is set by combining the online and offline share data of films such as "The Wandering Earth 2" on Maoyan Professional Edition.<sup>7</sup> The parameter  $\alpha = 0.3$  is based on the statistics of "incremental users" from the "China Network Audiovisual Development Research Report<sup>8</sup>". This offers a reliable empirical basis for the subsequent numerical analysis.

**Figs. 2–4** demonstrate how the optimal profits of content distributor, streaming platform and theater vary with  $v$  for two different  $\phi$  values ( $\phi = 0.2, 1$ ), but for the same values of the other parameters ( $\theta = 0.6, \gamma = 0.3, \beta = 0.5, \lambda = 0.7, k = 0.2, \alpha = 0.3$ ), respectively.

For the film distributor, the profitability of any single channel distribution strategy rises as  $v$  increases. The possible reason for this is that when a film is released through a single channel, more and more consumers are attracted to the film as their valuation increases. At this point the single release channel as a monopoly in the market is able to charge a higher price, so the distributor and distribution channel are possible to make higher profits.

The profitability of the film distributor in dual-channel distribution decreases with increasing  $v$ . This is because, with dual-channel releases, both the streaming platform and the theater are competing in the market, especially as consumers' valuations get higher. This has led to a drop in profits for film distributor. But when  $\phi$  is large if the film distributor chooses the TF strategy, the film distributor will gain higher profit. This is because at this time the utility loss of consumers deferring their purchases is small, so when a portion of consumers make purchases in the channel that was distributed first, thanks to the market expansion effect of the streaming platform, there will still be a portion of consumers purchasing in the streaming platform, and the streaming platform can gain more demand and profit, and thus the film distributor can also achieve higher profit.

Next, we analyze the optimal strategy for the film distributor. Observing **Fig. 2(a)** and **(b)** we obtain: when  $v$  is small, the optimal profit is the TF strategy; when  $v$  is large, the optimal strategy is the SO strategy. This is in line with industry practice, and we now observe an increasing number of films being released through streaming platforms on a single channel and earning high revenues. We analyze the optimal strategy choice problem for the film distributor by considering two scenarios in which the difference in release time has both a large and a small impact on consumers. The results show that the film distributor tends to earn higher profits through single-channel release, while benefiting from the market expansion effect of streaming platform, SO strategy profit is consistently higher than TO strategy.

For streaming platform, observing **Fig. 3(a)** and **(b)** we find that when  $v$  is low, streaming platform earns higher profit from the TF

<sup>6</sup> The information comes from <https://www.iimedia.cn/c400/106005.html/> (available at November 23, 2025).

<sup>7</sup> See <https://www.jiemian.com/article/10301571.html> / (available at November 20, 2025).

<sup>8</sup> See [https://www.sohu.com/a/878122348\\_121208683](https://www.sohu.com/a/878122348_121208683) / (available at November 28, 2025).

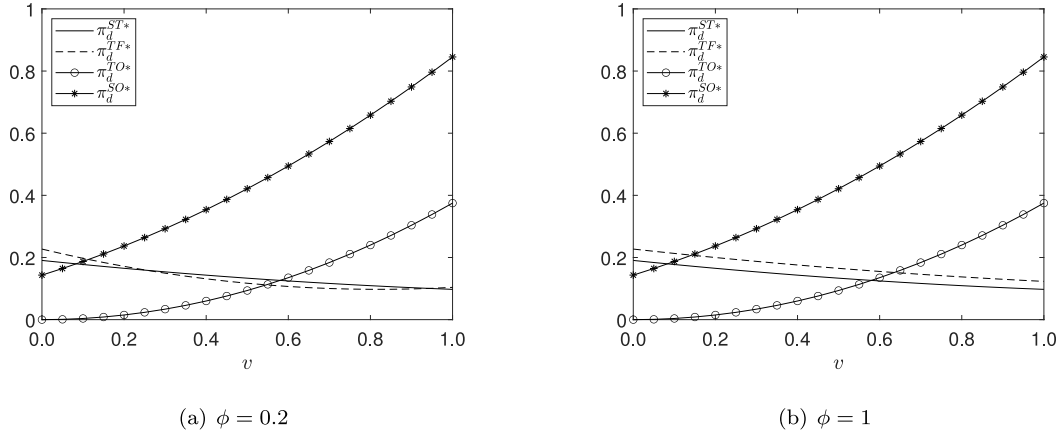


Fig. 2. Content distributor's profit change with consumers' valuation of the product for  $\theta = 0.6, \gamma = 0.3, \beta = 0.5, \lambda = 0.7, k = 0.2, \alpha = 0.3$ .

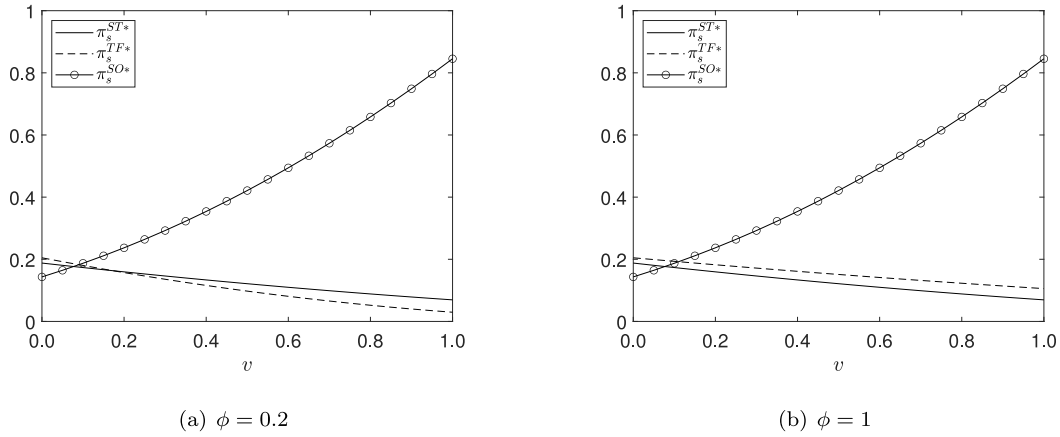


Fig. 3. Streaming platform's profit change with consumers' valuation of the product for  $\theta = 0.6, \gamma = 0.3, \beta = 0.5, \lambda = 0.7, k = 0.2, \alpha = 0.3$ .

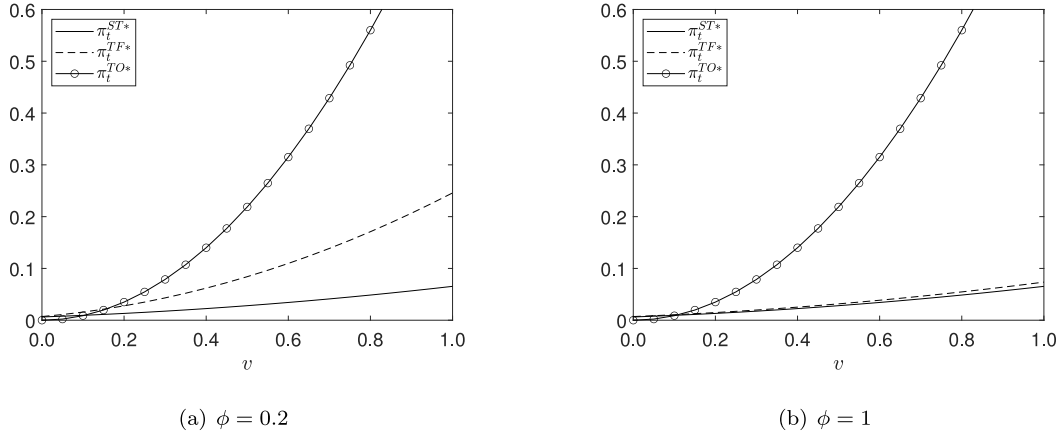


Fig. 4. Theater's profit change with consumers' valuation of the product for  $\theta = 0.6, \gamma = 0.3, \beta = 0.5, \lambda = 0.7, k = 0.2, \alpha = 0.3$ .

strategy; when  $v$  is high, profit is higher under the SO strategy. For the theater, whose profits consistently change in the opposite direction from the streaming platform when it comes to dual-channel releases. But that does not mean that the competition between them makes it impossible for them to be more profitable at the same time. Fig. 4(a) and (b) expose that: when  $v$  is small, the theater channel gets higher profit from the TF strategy; when  $v$  is large, it gets higher profit from the TO strategy. And we also find that when  $v$  is small, the streaming platform and theater simultaneously obtain higher profits from the TF strategy, regardless of the change of  $\phi$  value.

Figs. 5–7 show how the optimal profits of the film distributor, streaming platform, and theater vary with  $\theta$  for two different values of  $v$  ( $v=0.6, 1$ ), but for the same values of the other parameters ( $\phi = 0.6, \gamma = 0.3, \beta = 0.5, \lambda = 0.7, k = 0.6, \alpha = 0.5$ ), respectively.

Observing Fig. 5 we find that when the differences between channels become smaller, the profits of film distributors always increase except for the TO strategy. Among them, the SO strategy has the fastest profit growth. When the channel differences are small and the consumer valuation is high, the distributor's optimal strategy is SO strategy, otherwise, the optimal strategy is TF strategy.

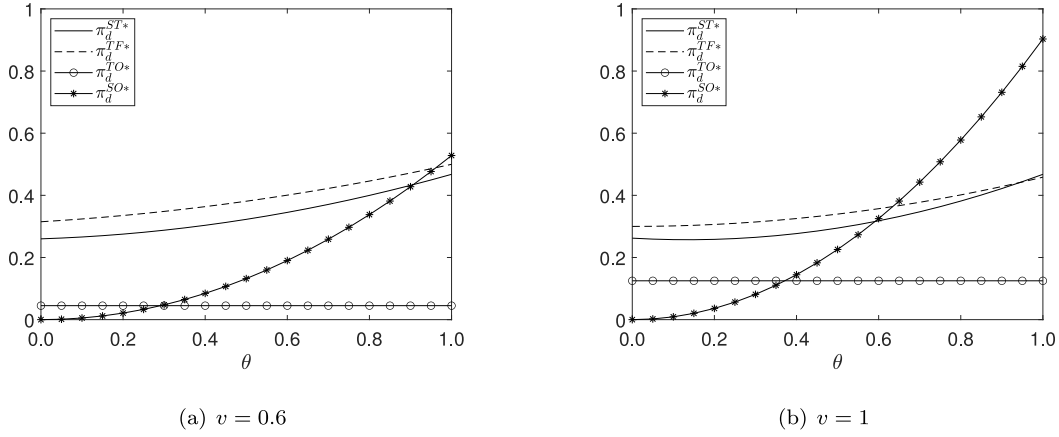


Fig. 5. Content distributor's profit change with channel differences for  $\phi = 0.6, \gamma = 0.3, \beta = 0.5, \lambda = 0.7, \alpha = 0.5, k = 0.6$ .

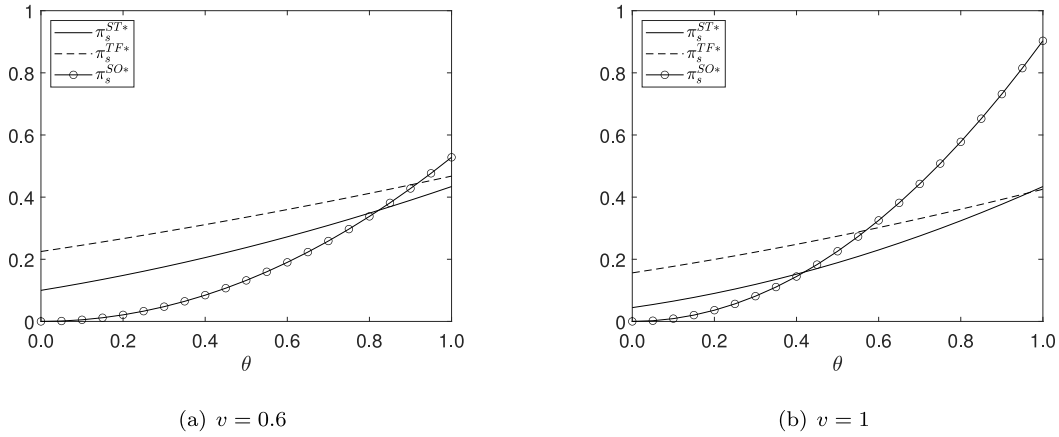


Fig. 6. Streaming platform's profit change with channel differences for  $\phi = 0.6, \gamma = 0.3, \beta = 0.5, \lambda = 0.7, \alpha = 0.5, k = 0.6$ .

For the streaming platform, the reduced differences between channels means that the platform is more competitive, which allows them to make a higher profit. Examining Fig. 6 reveals that: when consumer valuation is small, if  $\theta$  is small, the streaming platform has the highest profit under the TF strategy; if  $\theta$  is large, the platform has the highest profit under the SO strategy. If the consumer valuation is large, the most profitable strategy for the platform is the TF strategy when the value of  $\theta$  is small; when the value of  $\theta$  is medium, the most profitable strategy for the platform is the SO strategy.

For the theater, when the consumer valuation is small, the TF strategy is always the most profitable, regardless of channel differences. When consumer valuation is large, the optimal strategy is converted from the TF strategy to the TO strategy as the difference between channels decreases. At this time, consumers have a strong willingness to consume, and more profits can be gained for the theater by prioritizing distribution. However, as the difference between channels decreases, the competitiveness of the streaming platform increases, and in order to avoid the loss of consumers, exclusive distribution through theater channels can bring higher profits to theaters.

Fig. 8 illustrates how total demand varies with channel differences under four different strategies for two different values of  $\phi$  ( $\phi = 0.2, 1$ ), but for the same values of the other parameters ( $v = 0.6, \gamma = 0.3, \beta = 0.5, \lambda = 0.7, k = 0.7, \alpha = 0.3$ ), respectively.

By analyzing Fig. 8, we can see that benefiting from the platform expansion effect, the SO strategy has a much higher demand than the TO strategy as the difference between the dual channels decreases and is associated with obtaining the same demand as the dual-channel distribution at  $\theta = 1$ . It is understandable why more and more films are distributed only through streaming platforms. In addition, the total

demand of dual channel is always higher than the total demand of single channel and the total demand of TF strategy is higher than that of ST strategy, so it is the best for distributors who want to get more attention to release their new film through TF strategy, and this can also be found from the practice of the industry.

## 5. The welfare analysis

This section conducts a systematic comparison of the welfare levels under the four release strategies. Through comparing the social welfare among different strategies, we can more comprehensively assess the impact of each strategy on market participants and society as a whole. This, in turn, provides a reference for strategic decision-making by film distributors and policy formulation by regulatory authorities. Based on the equilibrium solutions obtained from the preceding analysis, we present the analytical expressions for the social welfare under each strategy as follows.

$$\begin{aligned}
 SW^{ST} &= \frac{(\theta(v + \lambda) - v)((v + \lambda)\theta(10 + 11\alpha) + v(\alpha - 10))}{72k} \\
 &\quad + \frac{\theta(v + \lambda)(5\alpha + 6) + v(6 + \alpha)}{12} - \frac{(2 + \alpha)k}{8}, \\
 SW^{TF} &= \frac{(\theta(e^{-b\tau}v + \lambda) - v)(\theta(e^{-b\tau}v + \lambda)(8\alpha + 7) - 7v)}{64k} \\
 &\quad + \frac{\theta(e^{-b\tau}v + \lambda)(40\alpha + 38) + 26v}{64} - \frac{17k}{64}, \\
 SW^{TO} &= \frac{3v^2}{8k}, \quad SW^{SO} = \frac{3(1 + \alpha)(\theta(v + \lambda))^2}{8k}.
 \end{aligned}$$

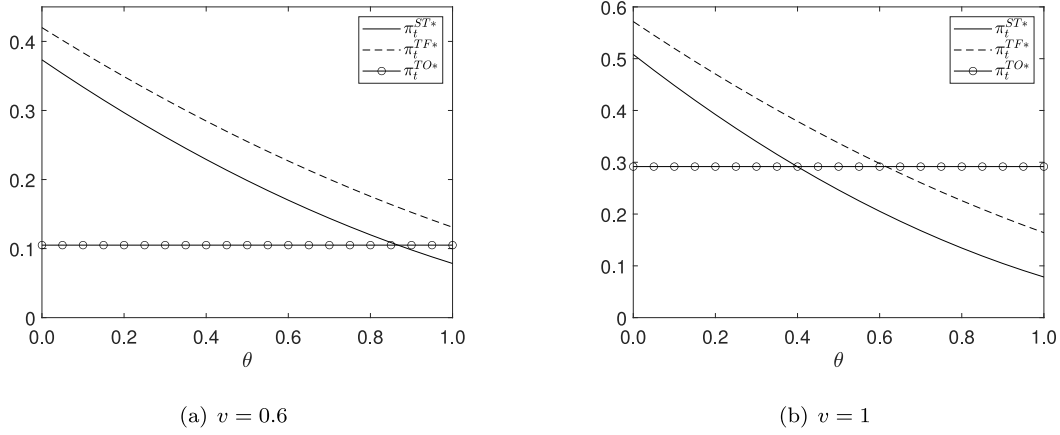


Fig. 7. Theater's profit change with channel differences for  $\phi = 0.6, \gamma = 0.3, \beta = 0.5, \lambda = 0.7, \alpha = 0.5, k = 0.6$ .

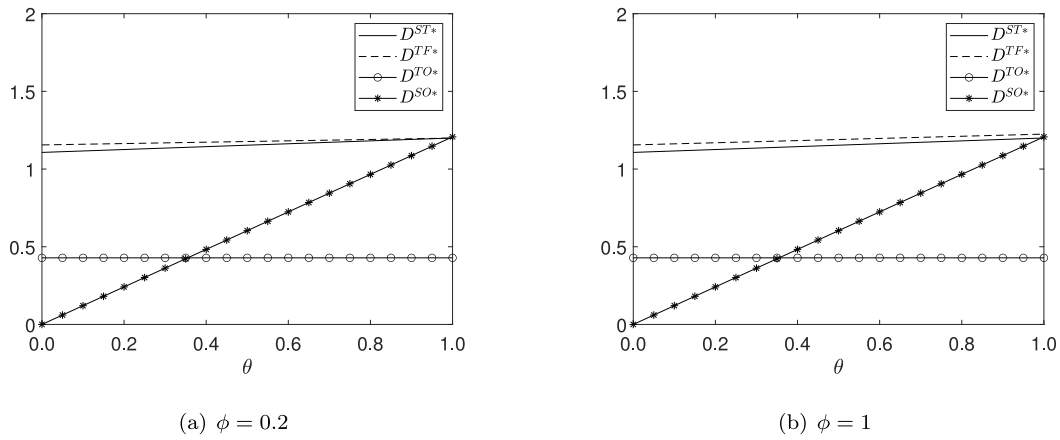


Fig. 8. Total demand varies with channel differences under four different strategies for  $v = 0.6, \gamma = 0.3, \beta = 0.5, \lambda = 0.7, k = 0.7, \alpha = 0.3$ .

We further compare the welfare levels among different distribution strategies. Specifically, in the case of single-channel release, the relative level of social welfare primarily depends on the degree of channel differentiation. [Proposition 5](#) formally presents the results of this comparison.

**Proposition 5.** *When the content distributor opts to release via a single channel, the social welfare under the TO strategy is higher than that under the SO strategy if and only if  $0 < \theta < \frac{v}{(v+\lambda)\sqrt{1+\alpha}}$ ; otherwise, the social welfare under the SO strategy is higher.*

[Proposition 5](#) indicates that under single-channel distribution strategies, when channel differentiation is substantial, the social welfare under the TO strategy exceeds that under the SO strategy. This is because of the significant difference in viewing experience between streaming platforms and theaters, which results in a considerable discount in consumer-perceived utility. In contrast, theaters hold an advantage in aggregate social welfare as the irreplaceable immersive viewing experience generates higher perceived utility per viewer. On the contrary, when channel differentiation is modest, the social welfare under the SO strategy is higher than that under the TO strategy. Streaming platforms have a viewing experience comparable to that of theaters, offer lower prices, and provide supplementary services. This combination attracts a significantly larger consumer base. As a result, the social welfare under the SO strategy surpasses that under the TO strategy. In summary, the social welfare under single-channel strategies depends on the degree of channel differentiation. Therefore, film distributors should strategically match channel selection with the extent to which a film depends on audiovisual immersion, and regulatory authorities are recommended to

promote benign competition between theaters and streaming platforms in the area of differentiated experiential offerings, thus facilitating the sustainable development of the film industry ecosystem.

Owing to the complexity of the expressions, we carry out a numerical analysis to further investigate the influence of channel differentiation on social welfare. [Fig. 9](#) demonstrates how social welfare changes with channel differentiation under different parameter values.

From [Fig. 9](#), it is evident that if the film distributor opts for a dual-channel distribution strategy, when the differences between channels are small, the social welfare under the TF strategy is higher than that under the ST strategy. Conversely, when the channel differences are large, the social welfare under the ST strategy is higher than that under the TF strategy. This is because, when the channel differences are small, the substitutability between the streaming platform and the theater intensifies. Consequently, the ST strategy triggers direct price competition between the two channels. Nevertheless, the intense competition results in a substantial decline in profits for both the theater and the streaming platform, leading to significant efficiency losses and, as a result, lower total social welfare. In contrast, through delaying the release, the TF strategy effectively mitigates the direct competitive conflict stemming from high channel substitutability, thus reducing efficiency loss and attaining higher social welfare.

Conversely, when the differences between channels are substantial, the utility discount related to the streaming platform becomes prominent, and the complementarity between the streaming platform and the theater intensifies. In this situation, the ST strategy effectively capitalizes on the advantages of the streaming platform, enticing users with differentiated experiences. Meanwhile, under the TF strategy, the

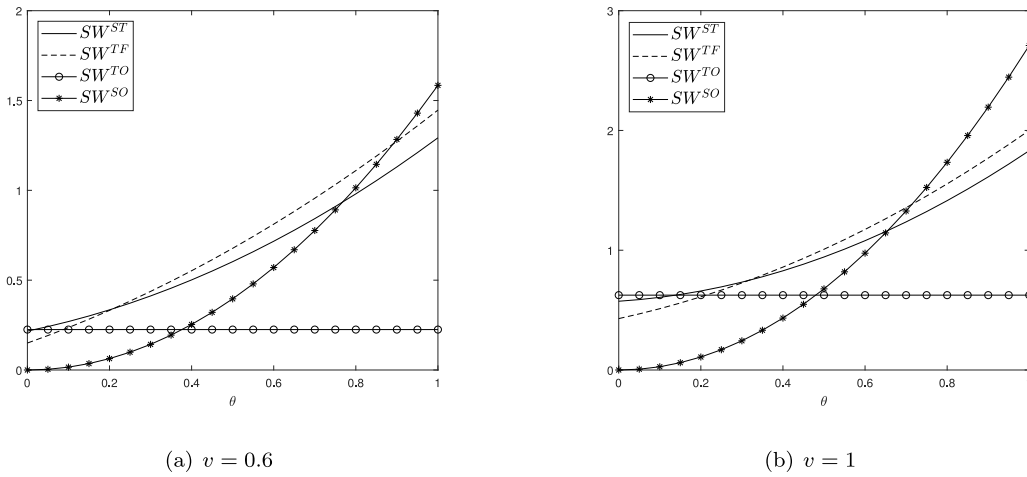


Fig. 9. Social welfare change with channel differences for  $\phi = 0.6, \gamma = 0.3, \beta = 0.5, \lambda = 0.7, \alpha = 0.5, k = 0.6$ .

delayed release causes a further shrinkage in demand on the streaming platform because of utility loss, and it fails to make up for the potential consumers lost during the early-release window, leading to relatively lower social welfare. In summary, channel differences modify the substitution-complementarity relationship between the two channels, thus resulting in the differentiated performance of social welfare under the ST and TF strategies. This implies that policymakers should formulate policies that promote vertical differentiation and curb pure price competition, thereby safeguarding the theater channel while fully maintaining the efficiency gains brought by the streaming platform.

## 6. Extensions

In this section, by relaxing specific assumptions or modifying certain settings, we extend our basic model to the following three scenarios. Through the analysis of these three cases, we aim to assess the robustness of our main findings and provide additional insights into the impact of film release channel choice in diverse contexts.

### 6.1. Consumer heterogeneity

In the primary model, we assume that consumers are homogeneous. However, in the real-world scenario, consumers display inherent differences in preference when it comes to channel selection. To account for this crucial characteristic, we introduce consumer heterogeneity by clearly dividing the market into two types: “theater-loyal” and “channel-balancing”. We then analyze how this structural disparity impacts the equilibrium strategies and profits of content distributors, streaming platforms, and theaters. We assume that the proportion of “theater-loyal” consumers in the market is  $\rho$  ( $0 < \rho < 1$ ), while the proportion of “channel-balancing” consumers is  $1 - \rho$ . For theater-loyal consumers, they only consider the theater channel. Their utility function is  $U_t$ , and they will purchase if and only if  $U_t > 0$ . For channel-balancing consumers, their behavior is fully consistent with the primary model. Based on this new market structure, we derive the equilibrium solutions for each strategy (See the Appendix for the equilibrium solutions). By analyzing the equilibrium solutions, we can draw the following conclusions that are closely related to management practices.

**Proposition 6.** *When the content distributor chooses to release through a single channel,*

(a) *If  $0 < \rho < \rho_1$ , the content distributor under the SO strategy gets higher profit; if  $\rho_1 < \rho < 1$ , the profit of content distributor under the TO strategy is higher;*

(b) *If  $0 < \rho < \rho_2$ , the demand for the SO strategy is higher than that for the TO strategy; if  $\rho_2 < \rho < 1$ , the demand for the TO strategy is higher;*

(c) *If  $0 < \rho < \rho_3$ , the profit of the streaming platform under the SO strategy is higher than that of the theater under the TO strategy,  $\pi_s^{SO*} > \pi_t^{TO*}$ ; otherwise,  $\pi_s^{SO*} < \pi_t^{TO*}$ .*

A low value of  $\rho$  suggests that the majority of consumers in the market are channel-balancing, they have no additional willingness to pay for the immersive viewing experience and place greater emphasis on convenience and price. In this situation, the streaming platform takes advantage of its market expansion effect and additional value-added services to achieve higher profitability than the theater. Meanwhile, the SO strategy is more profitable for the distributor than the TO strategy. Conversely, a high  $\rho$  implies the presence of a significant segment of theater-loyal consumers. Here, theaters utilize their distinctive and hard-to-replicate immersive viewing experience to establish a strong differentiated position. Under these circumstances, adopting the TO strategy enables content distributors to maximize both profit and demand. Meanwhile, it also allows theaters to generate higher channel profits than the streaming platform by leveraging their unique experiential advantages.

**Proposition 7.** *For any  $\rho \in (0, 1)$ , the profit relationship under different distribution strategies is as follows:*

- (a) *The streaming platform’s profit satisfies:  $\pi_s^{SO*} > \pi_s^{TF*} > \pi_s^{ST*}$ .*
- (b) *The theater’s profit satisfies:  $\pi_t^{ST*} > \pi_t^{TO*} > \pi_t^{TF*}$ .*

For the streaming platform, regardless of the value of  $\rho$ , the SO strategy always remains the most profitable. This stems from its capacity to eliminate direct price competition with the theater, exclusively appropriate the complete consumer surplus resulting from market expansion effects, and avoid the utility depreciation caused by release delays. In contrast, Proposition 7 demonstrates that theater achieves optimal profitability not through market exclusivity but through the ST strategy, where controlled competition actually strengthens its differentiated market position. This fundamental asymmetry reveals the underlying logic of channel competition: while the streaming platform reaches absolute profit optimum through full market exclusivity, the theater attains its profit maximum under limited channel coexistence through the ST strategy. These findings provide theoretical justification for streaming platforms’ pursuit of content exclusivity while simultaneously offering strategic validation for theaters’ participation in dual-channel distribution through experiential differentiation.

We conducted a detailed analysis of Fig. 10 to explore the influence mechanism of the proportion  $\rho$  on the profits of content distributor and systematically compared the results with those of the previous baseline model.

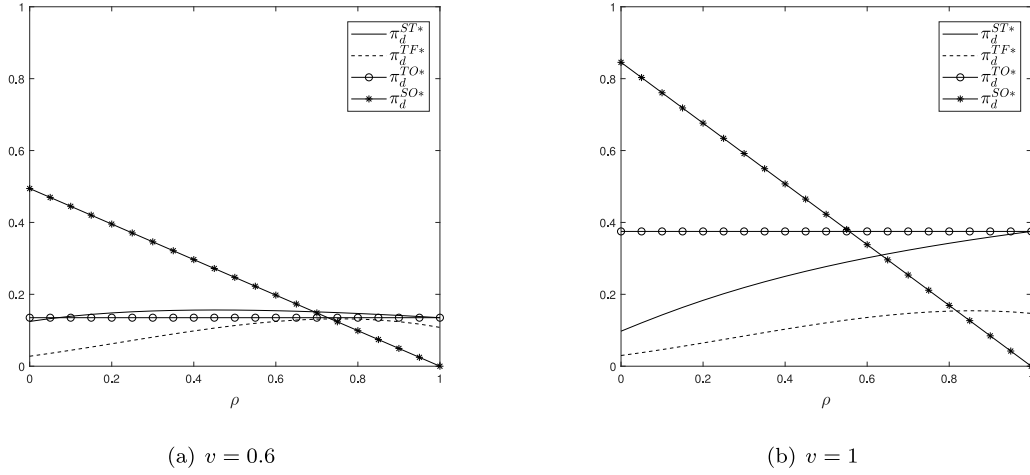


Fig. 10. Content distributor's profit change with preference differences for  $\theta = 0.6, \phi = 0.6, \gamma = 0.3, \beta = 0.5, \lambda = 0.7, \alpha = 0.5, k = 0.6$ .

Fig. 10 indicates that when  $\rho$  is large, the core contradiction in channel selection undergoes a shift. At this juncture, the perceived value of the film itself emerges as the crucial variable that determines the optimal strategy. If the value of the film is high (i.e.,  $v$  is large), which implies strong appeal, the distributor can fully exploit the differentiated viewing experience offered by the theater channel and the high willingness-to-pay of loyal audiences by adopting the TO strategy. This approach maximizes the content premium and prevents profit dilution across multiple channels. Conversely, if the film's value is relatively low (i.e.,  $v$  is small), even with a large number of theater-loyal users, its limited appeal is insufficient to sustain the revenue of a single theater channel. In such a case, the ST strategy should be selected. It meets the experience requirements of the core audience through the theater and, at the same time, leverages the low entry barrier and flexibility of the streaming platform to reach price-sensitive users. This helps to expand the audience scale and optimize the overall profit.

## 6.2. Nash bargaining

In the benchmark model, we assume that the royalty rates  $\gamma$  and  $\beta$  are exogenously given. However, in practical scenarios, such rates may also be determined through negotiations between distributors and channel parties. To address this limitation, we introduce the Nash Bargaining Game in the extended model to endogenize the royalty rates, thereby more accurately reflecting the strategic interaction between the two parties in profit sharing. We assume that the distributor and the channel party (theater or streaming platform) engage in negotiations before the film's release to determine the optimal royalty rates. Let the bargaining power of the distributor and the channel party be  $\delta_d$  and  $1 - \delta_d$  respectively. Under the Nash bargaining framework, the two parties determine the optimal royalty rates by maximizing the following objective function:

Under single-channel strategies, for the SO strategy, the negotiation occurs between the distributor and the streaming platform to determine the optimal royalty rate  $\beta$ . The two parties maximize the Nash bargaining product to determine the endogenous royalty rate  $\beta$ .

$$\max_{\beta} [\delta_d \ln(\pi_d^{SO} - 0) + (1 - \delta_d) \ln(\pi_s^{SO} - 0)]. \quad (11)$$

Similarly, for the TO strategy, the optimal royalty rate  $\gamma$  can be obtained:

$$\max_{\gamma} [\delta_d \ln(\pi_d^{TO} - 0) + (1 - \delta_d) \ln(\pi_t^{TO} - 0)]. \quad (12)$$

The following lemma presents the relationship between the endogenous royalty rate and the bargaining power of the two parties:

**Lemma 1.** Under single-channel strategies, the relationship between the endogenous royalty rate and the bargaining power of the two parties can be derived:  $\beta^* = \gamma^* = \delta_d$ .

Under dual-channel strategies, the Nash bargaining process is more complex compared to the single-channel case. For both strategies, we have well-defined profit functions (as shown in Table 1) that serve as the basis for bargaining. We incorporate these profit functions, along with the parties' disagreement payoffs in the event of negotiation breakdown, into the Nash bargaining framework. The specific settings are as follows:

When negotiations between the content distributor and the streaming platform break down, The distributor's disagreement payoff  $D_{d,s}$  is the inability to release the film via the streaming platform channel, leaving only the TO strategy model. In this case, the distributor's profit depends on the royalty rate  $\gamma$  negotiated with the theater under the single-channel arrangement,  $D_{d,s}(\gamma) = \pi_d^{TO}(\gamma) = \frac{\gamma v^2}{4k} = \frac{\delta_d v^2}{4k}$ . The streaming platform's disagreement payoff  $D_s$  is the inability to stream the film, resulting in  $D_s = 0$ . Similarly, when negotiations between the content distributor and the theater break down, the distributor's disagreement payoff is  $D_{d,t}(\beta) = \pi_d^{SO}(\beta) = \frac{\beta(1+\alpha)\theta^2(v+\lambda)^2}{4k} = \frac{\delta_d(1+\alpha)\theta^2(v+\lambda)^2}{4k}$ . The theater's disagreement payoff is  $D_t = 0$ .

Thus, under the ST strategy, the objective function for negotiations with the platform and the theater are:

$$\max_{\beta} [\delta_d \ln(\pi_d^{ST} - D_{d,s}) + (1 - \delta_d) \ln(\pi_s^{ST} - D_s)], \quad (13)$$

$$\max_{\gamma} [\delta_d \ln(\pi_d^{ST} - D_{d,t}) + (1 - \delta_d) \ln(\pi_t^{ST} - D_t)]. \quad (14)$$

By solving the first-order conditions, We obtain the relationship between the endogenous royalty rate and the bargaining power of the two parties:

**Lemma 2.** Under the ST strategy, the relationship between the endogenous royalty rate and the bargaining power of the two parties can be derived:

$$\beta^{ST*} = \frac{\delta_d[1-(1-\delta_d)K_1^{ST}]}{1-(1-\delta_d)^2K_1^{ST}K_2^{ST}}, \quad \gamma^{ST*} = \frac{\delta_d[1-(1-\delta_d)K_2^{ST}]}{1-(1-\delta_d)^2K_1^{ST}K_2^{ST}}.$$

The equilibrium solutions in this case are all long, and for ease of viewing and aesthetics, we use some letters to replace parts of them:  $K_1^{ST} = \frac{E^{ST}-H}{A^{ST}}$ ,  $K_2^{ST} = \frac{A^{ST}-F}{E^{ST}}$ ,  $A^{ST} = \frac{(1+\alpha)(3k-v+\theta v+\theta\lambda)^2}{18k}$ ,  $E^{ST} = \frac{(3k+v-\theta v-\theta\lambda)^2}{18k}$ ,  $H = \frac{v^2}{4k}$ ,  $F = \frac{(1+\alpha)\theta^2(v+\lambda)^2}{4k}$ .

Similarly, for the TF strategy, the objective function for negotiations with the platform and the theater are:

$$\max_{\beta} [\delta_d \ln(\pi_d^{TF} - D_{d,s}) + (1 - \delta_d) \ln(\pi_s^{TF} - D_s)], \quad (15)$$

$$\max_{\gamma} [\delta_d \ln(\pi_d^{TF} - D_{d,t}) + (1 - \delta_d) \ln(\pi_t^{TF} - D_t)]. \quad (16)$$

**Lemma 3.** Under the TF strategy, the relationship between the endogenous royalty rate and the bargaining power of the two parties can be derived:

$$\beta^{TF*} = \frac{\delta_d[1-(1-\delta_d)K_1^{TF}]}{1-(1-\delta_d)^2K_1^{TF}K_2^{TF}}, \gamma^{TF*} = \frac{\delta_d[1-(1-\delta_d)K_2^{TF}]}{1-(1-\delta_d)^2K_1^{TF}K_2^{TF}}.$$

Note that  $K_1^{TF} = \frac{E^{TF}-H}{A^{TF}}$ ,  $K_2^{TF} = \frac{A^{TF}-F}{E^{TF}}$ ,  $A^{TF} = \frac{(1+\alpha)(5k-v+\phi\theta v+\theta\lambda)^2}{32k}$ ,  $E^{TF} = \frac{(3k+v-\phi\theta v-\theta\lambda)^2}{16k}$ ,  $H = \frac{v^2}{4k}$ ,  $F = \frac{(1+\alpha)\theta^2(v+\lambda)^2}{4k}$ .

We establish that under single-channel strategies, the optimal royalty rate is a direct function of the distributor's bargaining power  $\delta_d$ . This result indicates that distributors with stronger bargaining power can obtain a larger share of channel revenues by negotiating for higher royalties, thus maximizing profits in exclusive distribution arrangements. This discovery is in line with industry practice, where leading distributors usually secure more favorable revenue-sharing terms in exclusive content licensing agreements. Under dual-channel strategies, the determination of the royalty rate follows a more intricate mechanism. The strategic interdependence among the film distributor, streaming platform, and theater implies that the outside options available when negotiations break down significantly influence the final royalty terms. This strategic dynamic suggests that royalty rates in dual-channel distribution are shaped not only by relative bargaining power but also by the level of inter-channel competition and market structure.

### 6.3. Dynamic influence of eWOM

This section introduces the Bass diffusion model and the electronic word-of-mouth (eWOM) effect to establish a dynamic multi-channel film distribution framework. In the context of film consumption, the initial audience affected by active marketing activities such as trailer releases and advertising campaigns forms the innovators in demand diffusion. Their decisions mirror the innovation effect in the Bass model. On the other hand, viewers influenced by social factors like friend recommendations and professional reviews are classified as imitators, and their behavioral logic represents the imitation effect in the Bass model. Considering the computational complexity, this section illustrates the computational procedures by taking the ST strategy as an example, and the details of other strategies are presented in Appendix.

Under the ST strategy, a dynamic optimization model is constructed based on the Bass Diffusion Model and eWOM effects. It is assumed that the total cumulative demand  $X^{ST}(t)$  for the film adheres to an extended Bass diffusion process, and its dynamic evolution is depicted by the following differential equation:

$$\begin{cases} \dot{X}^{ST}(t) = [e^{-ap_s^{ST}}\theta + e^{-ap_t^{ST}}] [pX^{ST}(t) + q] [1 - X^{ST}(t)], \\ X^{ST}(0) = 0. \end{cases} \quad (17)$$

Here,  $p_s^{ST}$  and  $p_t^{ST}$  denote the streaming platform price and the theater price, respectively;  $\theta$  represents the channel difference coefficient;  $p$  and  $q$  capture the imitation effect and innovation effect of eWOM, respectively;  $a$  represents the price sensitivity.

The profits of the film distributor, the streaming platform and theater are as follows:

$$\pi_d^{ST} = \int_0^T (\beta p_s^{ST} + \gamma p_t^{ST}) X^{ST}(t) dt, \quad (18)$$

$$\pi_s^{ST} = \int_0^T (1 - \beta) p_s^{ST} X^{ST}(t) dt, \quad (19)$$

$$\pi_t^{ST} = \int_0^T (1 - \gamma) p_t^{ST} X^{ST}(t) dt. \quad (20)$$

We can obtain the optimal prices,

$$p_s^{ST*} = \frac{1}{a} \left( 1 + \frac{1}{\beta} \ln \frac{[pX^{ST}(T) + q][1 - X^{ST}(T)]}{[pX^{ST}(t) + q][1 - X^{ST}(t)]} \right), \quad (21)$$

$$p_t^{ST*} = \frac{1}{a} \left( 1 + \frac{\eta}{\gamma} \ln \frac{[pX^{ST}(T) + q][1 - X^{ST}(T)]}{[pX^{ST}(t) + q][1 - X^{ST}(t)]} \right). \quad (22)$$

The parameter  $\eta$  is the channel relative importance coefficient, which reflects the relative contributions of the streaming platform and the theater in generating market demand.

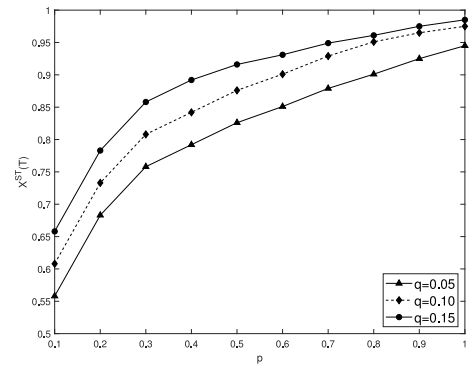


Fig. 11. Cumulative demand with the effect of eWOM and advertisement for  $a = 0.01, b = 0.005, \theta = 0.6, \phi = 0.6, \gamma = 0.3, \beta = 0.5, \lambda = 0.7, p = 0.6$ .

Fig. 11 depicts the dynamic changes in the cumulative demand for films under the influence of eWOM and advertising effects. Visually, the cumulative demand curve shows a characteristic S-shaped diffusion pattern. In the initial stages of release, the growth of cumulative demand is relatively gentle, mainly due to the adoption by innovators influenced by advertising and early marketing campaigns. As time elapses, the curve enters a period of rapid growth. In this period, imitators, affected by the social spread of eWOM, substantially speed up the accumulation of demand. Eventually, as the market nears saturation, the rate of demand growth decelerates, and the curve gradually levels off.

## 7. Conclusion

With the emergence and proliferation of streaming platforms, the practice of watching favored films via these platforms has gained widespread acceptance among consumers, leading to the increasing prevalence of releasing new films through streaming channels. Numerous research articles have delved into the issue of film release channels, exploring the implications and dynamics of this evolving distribution landscape. Based on this, our paper innovatively delves into the intricate interplay between streaming platform and theater channels through the development of a game-theoretic model. This model aims to ascertain the optimal release strategies for film distributors aiming to capitalize on both channels concurrently. In their quest for greater profitability, film distributors may strategically select their release channels. This study primarily examines the influence of market expansion effects, channel difference, and the declining consumer utility on the distribution strategies adopted by film distributors.

### 7.1. The main findings

This subsection summarizes the main findings of the study.

First, we analyze the optimal channel selection strategy of a film distributor when distributing a new film. Each strategy is evaluated by parametric analysis and numerical arithmetic, and we find that the TF and SO strategies perform better in terms of profits under certain conditions. These findings stand in stark contrast to the traditional view that releasing new films through the theater channel is the better way to go.

Second, longer release delays make the channel that releases first perform better in terms of profitability, while the channel that releases later always suffers a loss in profitability. Additionally, the reduction in differences between channels favors the streaming platform channel and hurts the profitability of the theater channel.

Third, when a film distributor chooses to release a new film through dual channels, the impact of the hassle cost coefficient allows the streaming platform and theater to be more profitable under different strategies. Both the streaming platform and theater make higher profits

under the ST strategy when the hassle cost factor for consumers is small. This is because both channels can more effectively segment consumers by establishing prices that foster moderate competition. In contrast, when the consumer's hassle cost coefficient is large, the streaming platform and theater prefer the TF strategy. When a film distributor chooses to release a new film through a single channel, the theater channel is more profitable under the TO strategy when the expansion effect is minimal, otherwise, the streaming platform is more profitable under the SO strategy.

Finally, when a film distributor adopts a dual-channel distribution strategy, the streaming platform can achieve higher demand and pricing under the ST strategy, while the theater channel can attain greater demand and pricing under the TF strategy, provided that the utility loss from consumers' delayed purchases is minimal. Conversely, when the utility loss from consumers delaying their purchases is significant, the streaming platform can achieve higher demand and pricing under the TF strategy, while the theater channel can attain greater demand and pricing under the ST strategy. When a film distributor chooses a single-channel distribution strategy, the TO strategy is priced higher than the SO strategy if channel differences are large, otherwise, the SO strategy is priced higher. Also, the demand for the TO strategy is greater than the demand for the SO strategy if channel differences are large, and higher for the SO strategy otherwise.

## 7.2. Managerial implications

This subsection provides actionable managerial implications from the perspectives of three key market participants: film distributors, streaming platforms, and theaters. These implications are elaborated as follows.

The analysis presented in this paper provides film distributors with profound insights. These insights can help them strike a delicate balance between overcoming channel-selection challenges and achieving profitability when interacting with channel managers. Specifically, when film distributors decide to release a new film, they must comprehensively consider various factors. Apart from royalty rates, it is essential to assess the market expansion impact of streaming platforms and the differences between traditional theatrical releases and streaming channels. By taking advantage of the strengths of streaming platforms, such as their fast dissemination speed, wide coverage, and high interactivity, distributors can effectively build a positive word-of-mouth for the new film, thus improving its overall market attractiveness. Additionally, the quality of film products is a crucial factor for film distributors when making decisions. Through diverse marketing strategies, including releasing film trailers, participating in film festivals, and organizing previews, they can increase the visibility of their films and understand consumer preferences. Choosing an appropriate release channel for new films has the potential to substantially boost profits in the film supply chain.

For streaming platform managers, maximizing the extensive influence of their platforms remains a top-priority objective. To achieve this, platforms frequently cooperate with one another to strengthen their expansionary impacts. Moreover, the unique characteristics of different channels are a crucial factor to consider. Considering the great significance of high-quality viewing experiences in today's streaming market, platforms have adjusted their business operations accordingly. For example, Tencent Video offers consumers 4 K picture quality and Dolby sound, and IQIYI provides complimentary VR glasses to improve the immersive viewing experience.

For theater managers, effectively countering the expansionary influence of streaming platforms requires a strategic approach. This entails closely monitoring box-office trends of films, cultivating strong relationships with film distributors, and adjusting screening strategies accordingly. Moreover, theaters can explore ways to offer diversified services that meet consumers' social and experiential needs, rather than

just focusing on the act of watching a film. By investing in experiential enhancements, such as high-end large-format screens (e.g., IMAX) and carefully curated social activities, theaters can directly boost the perceived value of their channel, thus strengthening their competitive position.

## 7.3. Future research

Our work also presents several opportunities for future research. First, our model focuses solely on the competitive cooperation between a single streaming platform and a single theater. In practice, film distribution may involve multi-homing behavior, suggesting that it would be valuable to investigate the competition and interaction dynamics among multiple streaming platforms and multiple theaters. Furthermore, a critical direction for future work lies in the empirical validation of our theoretical predictions. While our model provides clear testable hypotheses, testing these with real-world data would significantly strengthen the practical contribution of this research stream.

## CRedit authorship contribution statement

**Chi Zhou:** Writing – original draft, Funding acquisition. **Shujian Liu:** Writing – review & editing, Supervision, Software. **Dian Yao Kang:** Supervision, Methodology. **Zongxi Li:** Writing – review & editing, Investigation. **Yongjun Liu:** Writing – original draft.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix

**Proof of Section 3.1.** We simultaneously maximize the profits of the streaming platform and the theater to derive the equilibrium solution. Under the Scenario ST, the profits of the streaming platform and the theater are as follows:  $\pi_s^{ST} = (1 - \beta) \frac{(1+\alpha)(\theta(v+\lambda) - p_s^{ST} - v + p_t^{ST} + k)}{2k} p_s^{ST}$  and  $\pi_t^{ST} = (1 - \gamma) \left(1 - \frac{\theta(v+\lambda) - p_s^{ST} - v + p_t^{ST} + k}{2k}\right) p_t^{ST}$ .

To verify the existence of the optimal solution  $p_s^{ST}$ , we compute the second-order derivative of  $\pi_s^{ST}$  with respect to  $p_s^{ST}$ , i.e.,  $\frac{\partial^2 \pi_s^{ST}}{\partial (p_s^{ST})^2} = -\frac{(1+\alpha)(1-\beta)}{k} < 0$ . Then, the streaming platform's optimal price is characterized by the first-order condition on  $\pi_s^{ST}$ :  $\frac{\partial \pi_s^{ST}}{\partial p_s^{ST}} = \frac{(1+\alpha)(1-\beta)(k - 2p_s^{ST} + p_t^{ST} - v(1-\theta) + \theta\lambda)}{2k} = 0$ .

To establish the existence of optimal solution  $p_t^{ST}$ , we also compute the second-order derivative of  $\pi_t^{ST}$  with respect to  $p_t^{ST}$ , i.e.,  $\frac{\partial^2 \pi_t^{ST}}{\partial (p_t^{ST})^2} = -\frac{1-\gamma}{k} < 0$ . Then, the theater's optimal price is characterized by the first-order condition on  $\pi_t^{ST}$ :  $\frac{\partial \pi_t^{ST}}{\partial p_t^{ST}} = \frac{(1-\gamma)(k + p_s^{ST} - 2p_t^{ST} + v - \theta v - \theta\lambda)}{2k} = 0$ .

By jointly solving the first-order conditions of  $\frac{\partial \pi_s^{ST}}{\partial p_s^{ST}} = 0$  and  $\frac{\partial \pi_t^{ST}}{\partial p_t^{ST}} = 0$ , we easily obtain the optimal prices  $p_s^{ST*}$  and  $p_t^{ST*}$ . Substituting them into the profit functions and demand functions of the streaming platform and the theater, Section 3.1 is proven.

Utilizing the similar approach, we can validate the equilibrium solutions presented in Sections 3.2 through 3.4.

**Proof of Proposition 1.** By comparing profits of the streaming platform and theater under the two scenarios, we get the following results.

(a) According to Table 1, we have that  $\pi_s^{ST*} - \pi_s^{TF*} = \frac{(1+\alpha)(1-\beta)(16(3k-v(1-\theta)+\theta\lambda)^2 - 9(5k-v(1-\theta)+\theta\lambda)^2)}{288k}$ . Let  $\pi_s^{ST*} - \pi_s^{TF*} = 0$  to obtain  $k_1 = \frac{(4-3\phi)\theta v + \theta\lambda - v}{3}$ . Thus, if  $0 < k < k_1$ ,  $\pi_s^{ST*} > \pi_s^{TF*}$ , if  $k_1 < k < 1$ ,  $\pi_s^{ST*} < \pi_s^{TF*}$ .

(b) We derive conditions under which the theater's profit is greater through the ST strategy and TF strategy distribution. The proof is similar to (a), thus omitted here. The conditions are if  $0 < k < k_2$ ,  $\pi_t^{ST*} > \pi_t^{TF*}$ , if  $k_2 < k < 1$ ,  $\pi_t^{ST*} < \pi_t^{TF*}$ , where  $k_2 = \frac{(2\sqrt{2}-3)v + (3\phi-2\sqrt{2})\theta v + (3-2\sqrt{2})\theta\lambda}{9-6\sqrt{2}}$ .

**Proof of Proposition 2.** (a) Based on Table 2, we can get that  $D_s^{ST*} - D_s^{TF*} = \frac{(1+\alpha)(\theta\lambda-3k-v-(4+3\phi)\theta v)}{24k}$ . Let  $D_s^{ST*} - D_s^{TF*} = 0$  to obtain  $\phi_1 = \frac{\theta\lambda-3k-v}{3\theta v} + \frac{4}{3}$ . Thus, if  $0 < \phi < \phi_1$ ,  $D_s^{ST*} > D_s^{TF*}$ . Based on the similarity proof approach, we easily obtain when  $\phi_2 < \phi < 1$ ,  $D_t^{ST*} > D_t^{TF*}$ .  $\phi_2 = \frac{4}{3} - \frac{3k+2v-\theta\lambda}{3\theta v}$ . (b) Similarly, we can get when  $0 < \phi < \phi_1$ ,  $p_s^{ST*} > p_s^{TF*}$  and when  $\phi_3 < \phi < 1$ ,  $p_t^{ST*} > p_t^{TF*}$ .  $\phi_3 = \frac{2}{3} - \frac{\theta\lambda-3k-v}{3\theta v}$ .

**Proof of Proposition 3.** (a) From Table 1, we can obtain  $p_t^{TO*} - p_s^{SO*} = \frac{v-\theta(v+\lambda)}{2}$ . Set  $p_t^{TO*} - p_s^{SO*} > 0$ , we get  $0 < \theta < \frac{v}{v+\lambda}$ . Therefore, when  $0 < \theta < \frac{v}{v+\lambda}$ ,  $p_t^{TO*} > p_s^{SO*}$ , otherwise,  $p_t^{TO*} < p_s^{SO*}$ . (b) According to Table 2, we have that  $D^{TO*} - D^{SO*} = \frac{v-(1+\alpha)\theta(v+\lambda)}{2k}$ . Set  $D^{TO*} - D^{SO*} > 0$ , we get  $0 < \theta < \frac{v}{(1+\alpha)(v+\lambda)}$ . Thus, when  $0 < \theta < \frac{v}{(1+\alpha)(v+\lambda)}$ ,  $D^{TO*} > D^{SO*}$ , otherwise,  $D^{TO*} < D^{SO*}$ . (c) According to Table 1, we have that  $\pi_t^{TO*} - \pi_s^{SO*} = \frac{\gamma v^2 - (1+\alpha)\theta^2(v+\lambda)^2}{4k}$ . Set  $\pi_t^{TO*} - \pi_s^{SO*} > 0$ , we get  $\alpha < \frac{(1-\gamma)v^2}{(1-\beta)\theta^2(v+\lambda)^2} - 1$ . Thus, when  $\alpha < \frac{(1-\gamma)v^2}{(1-\beta)\theta^2(v+\lambda)^2} - 1$ ,  $\pi_t^{TO*} > \pi_s^{SO*}$ , otherwise,  $\pi_t^{TO*} < \pi_s^{SO*}$ .

**Proof of Proposition 4.** Next, from Table 1 and 2, we further derive the monotonicity of optimal prices, demands, and profits with respect to  $\tau$  and  $\theta$ . (a)  $\frac{\partial p_t^{TF*}}{\partial \tau} > 0$ ;  $\frac{\partial p_t^{TF*}}{\partial \tau} < 0$ ;  $\frac{\partial D_t^{TF*}}{\partial \tau} > 0$ ;  $\frac{\partial D_t^{TF*}}{\partial \tau} < 0$ ;  $\frac{\partial \pi_t^{TF*}}{\partial \tau} > 0$ ;  $\frac{\partial \pi_t^{TF*}}{\partial \tau} < 0$ . (b)  $\frac{\partial p_s^{ST*}}{\partial \theta} < 0$ ;  $\frac{\partial p_s^{ST*}}{\partial \theta} > 0$ ;  $\frac{\partial p_t^{TF*}}{\partial \theta} < 0$ ;  $\frac{\partial p_t^{TF*}}{\partial \theta} > 0$ ;  $\frac{\partial D_t^{TF*}}{\partial \theta} < 0$ ;  $\frac{\partial D_t^{TF*}}{\partial \theta} > 0$ ;  $\frac{\partial \pi_t^{TF*}}{\partial \theta} < 0$ ;  $\frac{\partial \pi_t^{TF*}}{\partial \theta} > 0$ ;  $\frac{\partial D_s^{ST*}}{\partial \theta} < 0$ ;  $\frac{\partial D_s^{ST*}}{\partial \theta} > 0$ ;  $\frac{\partial \pi_s^{ST*}}{\partial \theta} < 0$ ;  $\frac{\partial \pi_s^{ST*}}{\partial \theta} > 0$ .

**Proof of Section 5.** Social welfare under each strategy is defined as the sum of consumer surplus and the profits of the film distributor, the streaming platform, and the theater. Based on this, we derive the analytical expressions of social welfare for the four release strategies.

The consumer surplus under the four strategies is as follows:

$$CS^{ST} = \frac{(\theta(v+\lambda)-v)(\theta(v+\lambda)(7\alpha+2)+v(5\alpha-2))}{72k} + \frac{(\theta(v+\lambda)(6\alpha+1)+v(5\alpha+6))}{12} - \frac{5(2+\alpha)k}{8}.$$

$$CS^{TF} = \frac{(\theta(e^{-b\tau}v+\lambda)-v)(\theta(e^{-b\tau}v+\lambda)(6\alpha+1)+v(2\alpha-1))}{64k} + \frac{(\theta(e^{-b\tau}v+\lambda)(10\alpha+21)+v(11-10\alpha))}{64k} - \frac{(103+50\alpha)k}{64}.$$

$$CS^{TO} = \frac{v^2}{8k}, CS^{SO} = \frac{(1+\alpha)(\theta(v+\lambda))^2}{8k}.$$

Therefore, social welfare can be expressed as:  $SW^j = CS^j + \pi_d^j + \pi_s^j + \pi_t^j$ ,  $j \in \{ST, TF, TO, SO\}$ .

**Proof of Proposition 5.** According to the equilibrium solutions, we can easily get  $SW^{SO} - SW^{TO} = \frac{3((1+\alpha)\theta^2(v+\lambda)^2-v^2)}{8k}$ . Let  $SW^{SO} - SW^{TO} = 0$  to obtain  $\theta = \frac{v}{(v+\lambda)\sqrt{(1+\alpha)}}$ . Thus, if  $0 < \theta < \frac{v}{(v+\lambda)\sqrt{(1+\alpha)}}$ ,  $SW^{TO} > SW^{SO}$ , if  $\theta > \frac{v}{(v+\lambda)\sqrt{(1+\alpha)}}$ , then  $SW^{TO} < SW^{SO}$ .

**Proof of Section 6.1.** We simultaneously maximize the profits of the streaming platform and the theater to derive the equilibrium solution. Under the Scenario ST, the profits of the streaming platform and the

theater are as follows:  $\pi_s^{ST} = (1-\beta)(1-\rho)(1+\alpha) \left( \frac{\theta(v+\lambda)+p_s^{ST}-p_s^{ST}-v+k}{2k} \right) p_s^{ST}$  and  $\pi_t^{ST} = (1-\gamma)(\rho \frac{(v-p_s^{ST})}{k} + (1-\rho) \frac{(1-\theta(v+\lambda)+p_t^{ST}-p_s^{ST}-v+k)}{2k}) p_t^{ST}$ . To verify the existence of the optimal solution  $p_s^{ST}$ , we compute the second-order derivative of  $\pi_s^{ST}$  with respect to  $p_s^{ST}$ , i.e.,  $\frac{\partial^2 \pi_s^{ST}}{\partial (p_s^{ST})^2} = -\frac{(1+\alpha)(1-\beta)(1-\rho)}{k} < 0$ . Then, the streaming platform's optimal price is characterized by the first-order condition on  $\pi_s^{ST}$ :  $\frac{\partial \pi_s^{ST}}{\partial p_s^{ST}} = \frac{(1+\alpha)(1-\beta)(1-\rho)(k-2p_s^{ST}+p_t^{ST}-v+\theta(v+\lambda))}{2k} = 0$ .

To establish the existence of optimal solution  $p_t^{ST}$ , we also compute the second-order derivative of  $\pi_t^{ST}$  with respect to  $p_t^{ST}$ , i.e.,  $\frac{\partial^2 \pi_t^{ST}}{\partial (p_t^{ST})^2} = -\frac{(1-\gamma)(1+\rho)}{k} < 0$ . Then, the theater's optimal price is characterized by the first-order condition on  $\pi_t^{ST}$ :  $\frac{\partial \pi_t^{ST}}{\partial p_t^{ST}} = \frac{(1-\gamma)((1+\rho)(v-2p_t^{ST})-(1-\rho)\theta(v+\lambda)-k-p_s^{ST})}{2k} = 0$ . By jointly solving the first-order conditions of  $\frac{\partial \pi_s^{ST}}{\partial p_s^{ST}} = 0$  and  $\frac{\partial \pi_t^{ST}}{\partial p_t^{ST}} = 0$ , we easily obtain the optimal prices  $p_s^{ST*}$  and  $p_t^{ST*}$ . Substituting them into the profit functions and demand functions of the streaming platform and the theater.

Utilizing the similar approach, we can validate the equilibrium solutions presented in the remaining three models.

**The equilibrium solutions in Section 6.1.**

$$p_t^{ST*} = \frac{v+3k(1-\rho)-v\theta(1-\rho)-\theta\lambda(1-\rho)+3vp}{3+5\rho},$$

$$p_s^{ST*} = \frac{k(3+\rho)+\theta\lambda(1+3\rho)+v(-1+\theta-\rho+3\theta\rho)}{3+5\rho},$$

$$D_t^{ST*} = \frac{(1+\rho)(3k(1-\rho)-\theta\lambda(1-\rho)-v(-1+\theta-(3+\theta)\rho))}{2k(3+5\rho)},$$

$$D_s^{ST*} = \frac{(1-\rho)(1+\alpha)(k(3+\rho)+\theta\lambda(1+3\rho)+v(-1+\theta-\rho+3\theta\rho))}{2k(3+5\rho)},$$

$$\pi_t^{ST*} = \frac{(1-\gamma)(1+\rho)(v-v\theta+3k(1-\rho)-\theta\lambda(1-\rho)+v(3+\theta)\rho^2)}{2k(3+5\rho)^2},$$

$$\pi_s^{ST*} = \frac{(1+\alpha)(1-\beta)(1-\rho)(k(3+\rho)+\theta\lambda(1+3\rho)+v(-1+\theta-\rho+3\theta\rho))^2}{2k(3+5\rho)^2},$$

$$\pi_d^{ST*} = \frac{1}{2k(3+5\rho)^2}(\gamma(1+\rho)(v(1-\theta)+(3k-\theta\lambda)(1-\rho)+v(3+\theta)\rho^2+(1+\alpha)\beta(1-\rho)(k(3+\rho)+\theta\lambda(1+3\rho)+v(-1+\theta-\rho+3\theta\rho))^2),$$

$$p_t^{TF*} = \frac{v+3k(1-\rho)-\theta\lambda(1-\rho)+3vp-v\theta(1-\rho)\phi}{2+6\rho},$$

$$p_s^{TF*} = \frac{k(5+3\rho)+\theta\lambda(1+7\rho)+v(-1-3\rho+\theta\phi+7\theta\rho\phi)}{4+12\rho},$$

$$D_t^{TF*} = \frac{v+3k(1-\rho)-\theta\lambda(1-\rho)+3vp-v\theta(1-\rho)\phi}{8k},$$

$$D_s^{TF*} = \frac{(1+\alpha)(1-\rho)(k(5+3\rho)+\theta\lambda(1+7\rho)+v(-1-3\rho+\theta\phi+7\theta\rho\phi))}{8(k+3k\rho)},$$

$$\pi_t^{TF*} = \frac{(1-\gamma)(v+3k(1-\rho)-\theta\lambda(1-\rho)+3vp-v\theta(1-\rho)\phi^2)}{16(k+3k\rho)},$$

$$\pi_s^{TF*} = \frac{(1+\alpha)(1-\beta)(1-\rho)}{32(1+3\rho)^2}(k(5+3\rho)+\theta\lambda(1+7\rho)+v(-1-3\rho+\theta\phi+7\theta\rho\phi))^2,$$

$$\pi_d^{TF*} = \frac{1}{32} \left( \frac{2\gamma(v+3k(1-\rho)-\theta\lambda(1-\rho)+3vp-v\theta(1-\rho)\phi^2)}{k+3k\rho} + \frac{(1+\alpha)\beta(1-\rho)}{k(1+3\rho)^2}(k(5+3\rho)+\theta\lambda(1+7\rho)+v(-1-3\rho+\theta\phi+7\theta\rho\phi))^2 \right).$$

$$p_t^{TO*} = \frac{v}{2k}, D_t^{TO*} = \frac{v}{2k}, \pi_t^{TO*} = \frac{v^2(1-\gamma)}{4k}, \pi_d^{TO*} = \frac{v^2\gamma}{4k},$$

$$p_s^{SO*} = \frac{1}{2}\theta(v+\lambda), D_s^{SO*} = \frac{(1+\alpha)\theta(v+\lambda)(1-\rho)}{2k}, \pi_s^{SO*} = \frac{(1+\alpha)(1-\beta)(1-\rho)\theta^2(v+\lambda)^2}{4k},$$

$$\pi_d^{SO*} = \frac{(1+\alpha)(1-\rho)\beta\theta^2(v+\lambda)^2}{4k}.$$

**Proof of Proposition 6.** (a) According to the equilibrium solutions, we have that  $\pi_d^{TO*} - \pi_d^{SO*} = \frac{v^2\gamma+(1+\alpha)\beta\theta^2(v+\lambda)^2(-1+\rho)}{4k}$ . Let  $\pi_d^{TO*} - \pi_d^{SO*} = 0$  to obtain  $\rho_1 = 1 - \frac{v^2\gamma}{(v+\lambda)^2(1+\alpha)\beta\theta^2}$ . Thus, if  $0 < \rho < \rho_1$ ,  $\pi_d^{SO*} > \pi_d^{TO*}$ , if  $\rho_1 < \rho < 1$ ,  $\pi_d^{SO*} < \pi_d^{TO*}$ .

(b) According to the equilibrium solutions, we have that  $D^{TO*} - D^{SO*} = \frac{v-v\theta(1-\rho)-\theta\lambda(1-\rho)}{2k}$ . Let  $D^{TO*} - D^{SO*} = 0$  to obtain  $\rho_2 = 1 - \frac{v}{\theta(v+\lambda)}$ . Thus, if  $0 < \rho < \rho_2$ ,  $D^{SO*} > D^{TO*}$ , if  $\rho_2 < \rho < 1$ ,  $D^{SO*} < D^{TO*}$ .

(c) Similarly, we can get  $\pi_t^{TO*} - \pi_s^{SO*} = \frac{v^2(1-\gamma)-(1+\alpha)(1-\beta)\theta^2(v+\lambda)^2(1-\rho)}{4k}$ . Let  $\pi_t^{TO*} - \pi_s^{SO*} = 0$  to obtain  $\rho_3 = 1 - \frac{v^2(1-\gamma)}{(1+\alpha)(1-\beta)\theta^2(v+\lambda)^2}$ . Thus, if  $0 < \rho < \rho_3$ ,  $\pi_s^{SO*} > \pi_t^{TO*}$ , if  $\rho_3 < \rho < 1$ ,  $\pi_s^{SO*} < \pi_t^{TO*}$ .

**Proof of Proposition 7..** (a) We can easily get  $\pi_s^{SO*} - \pi_s^{TF*} = \frac{(1+\alpha)(1-\beta)(1-\rho)(8\theta^2(v+\lambda)^2 - \frac{(k(5+3\rho)+\theta\lambda(1+7\rho)+v(-1-3\rho+\theta\phi+7\theta\rho\phi))^2}{(1+3\rho)^2})}{32k}$ . We can obtain  $8\theta^2(v+\lambda)^2 - \frac{(k(5+3\rho)+\theta\lambda(1+7\rho)+v(-1-3\rho+\theta\phi+7\theta\rho\phi))^2}{(1+3\rho)^2} > 0$ , and

we can also obtain a denominator greater than 0,  $(1+\alpha)(1-\beta)(1-\rho) > 0$ , so  $\pi_s^{SO*} > \pi_s^{TF*}$ . Similarly, we can get  $\pi_s^{TF*} - \pi_s^{ST*} > 0$ , so  $\pi_s^{TF*} > \pi_s^{ST*}$ .

(b) According to the equilibrium solutions, we can easily get

$\pi_t^{ST*} - \pi_t^{TO*} = \frac{(-1+\gamma)}{4k} \left( v^2 - \frac{2(1+\rho)(v+3k(1-\rho)-v\theta(1-\rho)-\theta\lambda(1-\rho)+3v\rho)^2}{(3+5\rho)^2} \right)$ . We can obtain a denominator greater than 0,  $-1+\gamma < 0$ , and  $v^2 - \frac{2(1+\rho)(v+3k(1-\rho)-v\theta(1-\rho)-\theta\lambda(1-\rho)+3v\rho)^2}{(3+5\rho)^2} < 0$ . So  $\pi_t^{ST*} > \pi_t^{TO*}$ . Similarly, we can get  $\pi_t^{TO*} > \pi_t^{TF*}$ .

**Proof of Section 6.2.** As can be seen from Table 1,  $\pi_d^{ST*} = \frac{\beta(1+\alpha)(3k-v+\theta v+\theta\lambda)^2}{18k} + \frac{\gamma(3k+v-\theta v-\theta\lambda)^2}{18k}$ ,  $\pi_d^{TO*} = \frac{\gamma v^2}{4k}$ . Under strategy ST, let  $A = \delta_d \ln(\pi_d^{ST} - D_{d,s}) + (1 - \delta_d) \ln(\pi_s^{ST} - D_s)$ ,  $B = \delta_d \ln(\pi_d^{ST} - D_{d,t}) + (1 - \delta_d) \ln(\pi_s^{ST} - D_t)$ . By jointly solving the first-order conditions of  $\frac{\partial A}{\partial \beta} = 0$  and  $\frac{\partial B}{\partial \gamma} = 0$ , we easily obtain the optimal royalty rates  $\beta^*$  and  $\gamma^*$ .

Utilizing the similar approach, we can validate the equilibrium solutions presented in the remaining three models.

**Proof of Section 6.3.** (a) Under the ST strategy, to solve the profit maximization problem of the film distributor, the Hamiltonian function is introduced:

$$H^{ST} = \left[ \beta p_s^{ST} \theta \exp(-ap_s^{ST}) + \lambda_s(t) \theta \exp(-ap_s^{ST}) \right] [pX^{ST}(t) + q][1 - X^{ST}(t)] + \left[ \gamma p_t^{ST} \exp(-ap_t^{ST}) + \lambda_t(t) \exp(-ap_t^{ST}) \right] [pX^{ST}(t) + q][1 - X^{ST}(t)].$$

Here,  $\lambda_s(t)$  and  $\lambda_t(t)$  are the costate variables, representing the shadow prices of marginal value for the streaming channel and the theater channel respectively.

According to the Pontryagin's maximum principle, the optimal price path satisfies the first-order conditions:

$$\frac{\partial H^{ST}}{\partial p_s^{ST}} = [\beta(1 - ap_s^{ST}) - a\lambda_s(t)] \exp(-ap_s^{ST}) \theta [pX^{ST}(t) + q][1 - X^{ST}(t)] = 0,$$

$$\frac{\partial H^{ST}}{\partial p_t^{ST}} = [\gamma(1 - ap_t^{ST}) - a\lambda_t(t)] \exp(-ap_t^{ST}) [pX^{ST}(t) + q][1 - X^{ST}(t)] = 0.$$

The optimal pricing strategies are:  $p_s^{ST*} = \frac{1}{a} + \frac{\lambda_s(t)}{\beta}$ ,  $p_t^{ST*} = \frac{1}{a} + \frac{\lambda_t(t)}{\gamma}$ .

The costate variables  $\lambda_s(t)$  and  $\lambda_t(t)$  satisfy the adjoint equations,

$$\frac{d\lambda_s}{dt} = -\frac{\partial H^{ST}}{\partial X^{ST}}, \quad \frac{d\lambda_t}{dt} = -\frac{\partial H^{ST}}{\partial X^{ST}}.$$

Through integral transformation and the transversality condition  $\lambda_s(T) = \lambda_t(T) = 0$ , the analytical expressions of the costate variables are derived

$$\lambda_s(t) = \frac{1}{a} \ln \frac{[pX^{ST}(T) + q][1 - X^{ST}(T)]}{[pX^{ST}(t) + q][1 - X^{ST}(t)]}, \quad \lambda_t(t) = \eta \cdot \lambda_s(t).$$

Where  $\eta$  is the channel relative importance coefficient, reflecting the relative contributions of the streaming platform and the theater in market demand generation.

We can obtain the optimal price,

$$p_s^{ST*} = \frac{1}{a} \left( 1 + \frac{1}{\beta} \ln \frac{[pX^{ST}(T) + q][1 - X^{ST}(T)]}{[pX^{ST}(t) + q][1 - X^{ST}(t)]} \right),$$

$$p_t^{ST*} = \frac{1}{a} \left( 1 + \frac{\eta}{\gamma} \ln \frac{[pX^{ST}(T) + q][1 - X^{ST}(T)]}{[pX^{ST}(t) + q][1 - X^{ST}(t)]} \right).$$

(b) Under the SO strategy,  $X^{SO}(t) = \exp^{-ap_s^{SO}} \theta [pX^{SO}(t) + q][1 - X^{SO}(t)]$ .

(c) Under the TO strategy,  $X^{TO}(t) = \exp^{-ap_t^{TO}} [pX^{TO}(t) + q][1 - X^{TO}(t)]$ .

The profit functions for these three strategies are:  $\pi_d = \int_0^T (\beta p_s + \gamma p_t) X(t) dt$ ,  $\pi_s = \int_0^T (1 - \beta) p_s X(t) dt$ ,  $\pi_t = \int_0^T (1 - \gamma) p_t X(t) dt$ .

(d) Under the TF strategy, the distribution process is divided into two phases. In the first phase ( $0 < t < \tau$ ), only the theater channel is open, and the streaming platform has not yet released the film. Therefore, the online sales rate is  $X^{TF}(t) = \exp^{-ap_t^{TF}} [pX^{TF}(t) + q][1 - X^{TF}(t)]$ . In the second phase ( $t > \tau$ ), the streaming platform begins its release, entering a dual-channel model. At this point, the online sales rate is  $X^{TF}(t) = [\theta \phi \exp^{-ap_s} + \exp^{-ap_t^{TF}}] [pX^{TF}(t) + q][1 - X^{TF}(t)]$ . Therefore, the

profits of the film distributor, streaming platform, and theater are as follows:  $\pi_d^{TF} = \int_0^\tau \gamma p_t^{TF} X^{TF}(t) dt + \int_\tau^T (\beta p_s^{TF} + \gamma p_t^{TF}) X^{TF}(t) dt$ ,  $\pi_t^{TF} = \int_0^T (1 - \gamma) p_t^{TF} X^{TF}(t) dt$ ,  $\pi_s^{TF} = \int_\tau^T (1 - \beta) p_s^{TF} X^{TF}(t) dt$ .

## Data availability

No data was used for the research described in the article.

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